

Project Technical Report for the IREC

Team 605 Project Technical Report for the 2026 IREC

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This technical report presents a comprehensive overview of the Ozyman project, covering all phases from initial design decisions to final testing and results. It documents the contributions of the aerostructure, payload, and avionics divisions. Beyond describing the technical outcomes, this report aims to serve as a reference for future generations of the GAUL organization by detailing the methodologies, design rationales, and decision-making processes that guided the project.

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I Nomenclature

A = amplitude of oscillation
a = cylinder diameter
C_p = pressure coefficient
C_x = force coefficient in the x direction
C_y = force coefficient in the y direction
c = chord
dt = time step
F_x = X component of the resultant pressure force acting on the vehicle
F_y = Y component of the resultant pressure force acting on the vehicle
f, g = generic functions
h = height
i = time index during navigation
j = waypoint index
K = trailing-edge (TE) nondimensional angular deflection rate

II Introduction

The Aerospace Group from Laval University (GAUL, acronym in French) is a student-led organization based at Université Laval in Quebec City, Canada. The team brings together students from various disciplines, primarily Engineering Physics, Mechanical Engineering, and Electrical and Computer Engineering, while also including members from fields such as administration and the arts. This multidisciplinary structure enables the integration of technical expertise with project management and outreach capabilities.

This report presents the work carried out by the team on the *Ozyman* project. *Ozyman* is a two-stage high-power rocket designed to reach an altitude of 45,000 feet. The vehicle integrates a payload consisting of a deployable rover and a spectral camera, along with onboard imaging systems.

The report begins with an overview of the system architecture, followed by detailed descriptions of the propulsion, aerostructure, recovery, avionics, and payload subsystems. The Mission Concept of Operations (CONOPS) is then presented, outlining the planned flight sequence and operational procedures. Finally, the report concludes with a discussion of the key lessons learned throughout the development process. Supporting data, analyses, and additional documentation are provided in the appendices.

II.A Team Structure

GAUL is organized into four main technical divisions: Aerostructure, Avionics, Propulsion, and Payload, supported by a dedicated marketing team responsible for communications, partnerships, and funding. The President and Vice President oversee the overall direction of the team, ensuring coordination between all activities and alignment with project objectives. Supporting this leadership, the Treasurer and Marketing Director are responsible for financial management, sponsorship acquisition, and external communications, ensuring the sustainability of the project.

On the technical side, the Rocket Director is responsible for system-level integration and overall project execution, while the Technical Director ensures engineering consistency and validation across all subsystems. The technical work is divided into four main divisions each led by a dedicated team lead responsible for the design, development, and validation of their subsystem.

This structure promotes clear responsibility, efficient coordination, and strong collaboration between teams, which are essential for the successful development of a two-stage high-power rocket.

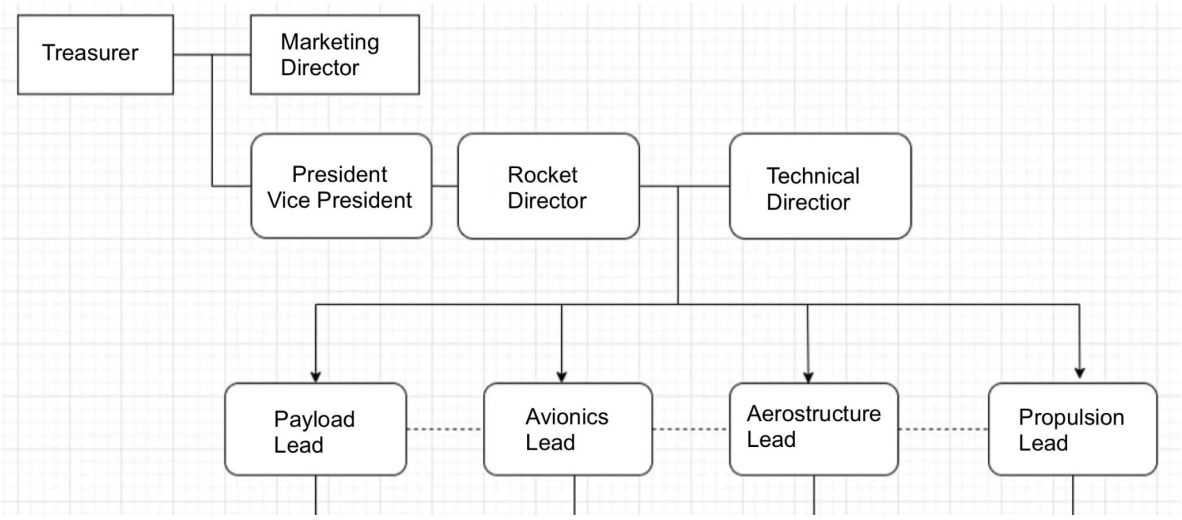


Fig. 1 Team Structure

III System Architecture Overview

III..1 General overview

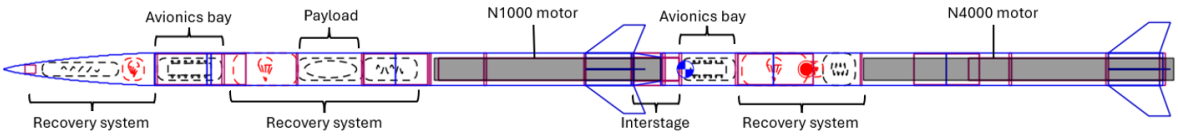


Fig. 2 Ozyman design overview

This two-stage rocket, targeting an altitude of 45 kft, has a diameter of 142 mm (5.6 in) to accommodate the CubeSat payload format while remaining as light and aerodynamic as possible. Both motors are 98 mm Aerotech reloads, assembled with centering rings and thrust transfer components inside the larger-diameter body tubes. Both stages use separate parachutes. The sustainer includes two independent recovery bays with two separation events, whereas the booster uses a single bay. Two of the three total recovery separations are actuated using pyrotechnic black powder charges, whereas sustainer apogee deployment uses an SRAD CO system for improved reliability in low-density atmosphere. A specific feature of the sustainer main parachute deployment is that the same separation event also releases the deployable payload, with the harness extracting it from the body tube. Stage separation is primarily drag-based, with a backup pyrotechnic charge controlled by the booster to improve reliability. The interstage coupling is achieved with the sustainer motor serving as the only connecting element, centered and retained on the interstage subsystem fixed to the top of the booster. Sustainer motor ignition is managed by a Blue Raven located in the sustainer avionics bay, with an igniter routed along the exterior of the motor and re-entering from the bottom end.

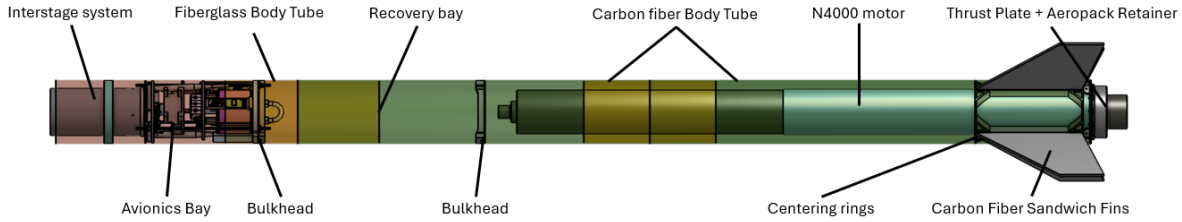


Fig. 3 Booster design overview

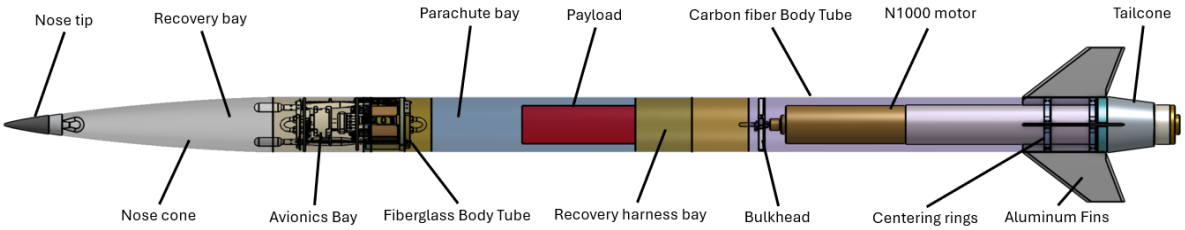


Fig. 4 Sustainer design overview

III..2 Design and Simulation Procedures

The flight performance predictions are entirely based on simulations and calculations, as no test flight will be performed prior to the competition flight. These simulations are mainly conducted with OpenRocket throughout the year, as both the design and the corresponding model evolve continuously and can be updated efficiently. However, OpenRocket is not the only software used for flight predictions.

To increase confidence in the results and potentially improve prediction accuracy, the same rocket was also modeled in RasAero, RocketPy, and our internally developed method named SimulOutil. Using four different simulation methods may appear redundant, but this redundancy is precisely what justifies their use. In past projects involving complex flights such as two-stage rockets, prediction discrepancies were encountered where OpenRocket and RasAero produced significantly different results, with no clear indication of which result should be trusted.

The addition of RocketPy provides the opportunity not only to validate the predicted apogee from a different mathematical approach, but also to improve flight dynamics predictions through three-dimensional trajectory simulation, among other capabilities. The fourth method, based on mathematics developed internally, also provides an apogee prediction for cross-validation while intentionally remaining as simple as possible. This simplicity, combined with full understanding of the model, is valuable to assess which predictions are the most physically realistic and to easily adjust or test calculation parameters.

More specifically, the developed method determines the acceleration at each time step of the flight, then obtains velocity and position through time integration. The acceleration at any instant is found by summing the three force components acting on the rocket and dividing by the vehicle mass at that instant, according to Newton's second law:

$$\sum F = ma \tag{1}$$

$$\frac{T - D - W}{m} = a \tag{2}$$

$$v = \int a \, dt \tag{3}$$

$$x = \int v \, dt \tag{4}$$

Following this simple mathematical framework, most of the effort is focused on determining the force components, which represent the real unknowns in this problem. Thrust data is taken directly from the motor thrust curve, while significant effort is dedicated to predicting drag and mass.

For drag, the primary approach is detailed vehicle modeling in multiple software tools to determine drag characteristics as a function of flight regime. Three of these tools are COTS software: OpenRocket, RasAero, and AeroLab. A fourth method is an internally developed analytical calculation code based on a DATCOM paper presenting empirical equations for predicting the aerodynamic drag of a small rocket. In addition, CFD simulations using Star-CCM+ are performed for specific configurations and flight regimes, as detailed in the appendix.

Finally, the rocket mass evolves significantly throughout the year as components are progressively designed. Initial estimates are generally made conservatively using expected dimensions and materials, then verified using previous rocket archives and increased slightly. Applying reasonable mass margins to each component helps compensate for small unforeseen additions and significantly reduces the risk of seeing performance predictions decrease throughout the year without sufficient margin to recover elsewhere.

III.A Aerostructure Subsystems

III.A.1 Airframe and couplers

Requirement	Description
RF transparency	Selected tubes must use radiotransparent material
Mass efficiency	Minimize tube mass across all sections
Buckling & bending	Resist structural loads under acceleration
Bearing pressure	Bolt holes must withstand parachute deployment and motor thrust loads

Design

The airframe consists of four SRAD body tubes, all with an inner diameter of 142 mm. This dimension was driven by the CubeSat payload, since a standard CubeSat has a diagonal of 141.4 mm, leaving minimal but sufficient clearance while keeping the frontal area, and thus aerodynamic drag, as small as possible. The nosecone is also SRAD and made with an exterior mold graciously produces by a local compagny called Elite Composite. It is a 5:1 Von Karman nosecone. This shape was chosen because it has a lower drag at supersonic velocities. With a cfd simulation like the one shown in figure, we can see the shock wave it produces.

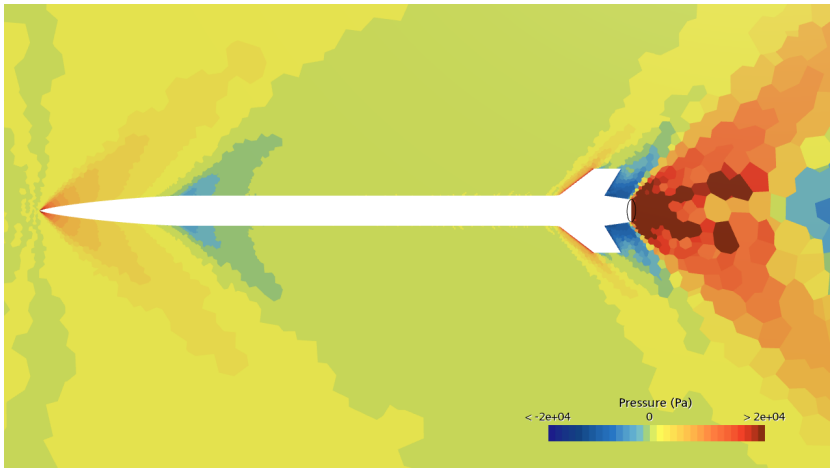


Fig. 5 Pressure scene of the sustainer at max velocity from a CFD simulation on Starr CCM+

Material selection follows a radio-transparency constraint: tubes housing avionics bays require unobstructed RF transmission for telemetry and payload communication, and are therefore fabricated from fiberglass (320 g/m² plain weave E-glass). All other tubes are fabricated from carbon fiber (3k 2×2 twill, 205 g/m²) to maximize stiffness-to-mass ratio. All four tubes share a common layup of $[90_5]_T$, so five plies with fibers with the tube axis (i.e. axial direction), which prioritizes strength to resist compression forces and bending moments. Since it's a woven fabric there is also the 0 degree fibers in the hoop direction. The properties of each tube are summarized in Table 1. All the tubes are student made by resin infusion on an aluminum mandrel with an oven cure. The method is exposed in appendix V.

Tube	Stage	Material	Layup	t (mm)	L (mm)	RF transparent
Nosecone	Sustainer	E-glass 320 g plain weave	$[0_5]_T$	1.6	700	Yes
Avionics & payload	Sustainer	E-glass 320 g plain weave	$[90_5]_T$	1.6	850	Yes
Motor	Sustainer	CF 3k 2×2 twill 205 g	$[90_5]_T$	1.5	1100	No
Avionics	Booster	E-glass 320 g plain weave	$[90_5]_T$	1.6	520	Yes
Motor	Booster	CF 3k 2×2 twill 205 g	$[90_5]_T$	1.5	1800	No

Table 1: Body tube specifications. All tubes have an inner diameter of 142 mm.

Coupler	Stage	Material	Layup	t (mm)	L (mm)	RF transparent
Avionics	Sustainer	E-glass 320 g plain weave	$[90_5]_T$	1.6	420	Yes
Motor	Sustainer	CF 3k 2×2 twill 205 g	$[90_5]_T$	1.5	300	No
Interstage	Booster	CF 3k 2×2 twill 205 g	$[90_7]_T$	1.8	215	No
Avionics	Booster	CF 3k 2×2 twill 205 g	$[90_5]_T$	1.6	350	No
Motor	Booster	CF 3k 2×2 twill 205 g	$[90_5]_T$	1.5	300	No

Table 2: Coupler specifications. All couplers except interstage have an inner diameter of 138 mm.

Analysis

The calculation method for the analysis is exposed in appendix V. This method can calculate the stress on a certain element of composite. By iterating along the length, we can get a finite element analysis to extract the safety factor of the airframe at each point. The result is shown in figure 6. As said in the appendix, we traction tested our composite in laboratory to get the most acurate calculation. The results of this testing, which are the engineering properties of our materials are shown in section V.

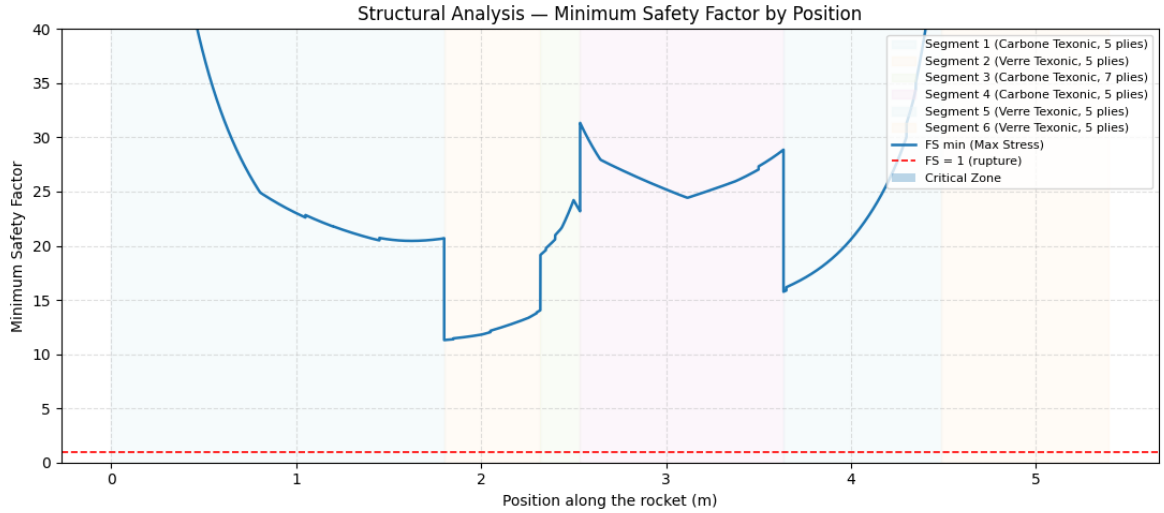


Fig. 6 Minimum safety factor along the length of the rocket due to motor thrust and aerodynamic loading

III.A.2 Bulkheads

Bulkheads serve as rigid structural interfaces within the rocket's internal structure, transferring mechanical loads between subsystems. Their main functions are anchoring parachute shock cords during recovery deployment, securing avionics and payload components to the fuselage, and transmitting motor thrust to the airframe. To fulfill these roles, the bulkheads must satisfy the following requirements

Requirements Each bulkhead must withstand the high mechanical loads generated during flight events, particularly parachute deployment, while remaining as lightweight as possible. The corresponding loads transmitted through the recovery harnesses are detailed in Table 9.

Design All bulkheads (fig. 25, 27, 29, 31, 33 and 35) are machined from 6061-T6 aluminium, with an outer diameter of 142 mm for the booster and 138 mm for the sustainer to match their respective fuselage internal diameters, and are secured to the fuselage by four radial M5 screws. Parachute cords are anchored through 3/8" U-bolts, except on the sustainer motor bulkhead, which uses a central eye-bolt that also connects to the motor casing. To minimize mass while resisting peak bending stresses, the bulkheads feature I-shaped or X-shaped internal ribs, resulting in masses ranging from 150 g for the sustainer upper avionics bay bulkhead to 243 g for the booster motor bulkhead. This mass optimization is critical to reach the 45,000 ft target apogee, as every gram saved on structural components directly contributes to the achievable altitude.

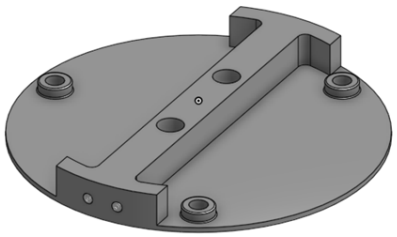


Fig. 7 BOOSTER - Avionic bay bottom bulkhead (as an example)

Analysis The mechanical analysis of each bulkhead followed a two-step approach. A first iteration based on the beam theory was used to size the components and identify the dimensioning load cases, followed by a finite element analysis in Siemens NX Nastran to validate the final design under realistic boundary conditions. The radial M5 fasteners were sized using threaded insert calculations from Shigley's, considering shear, bearing and tensile yielding failure modes. Section V in the appendix presents the detailed calculations and FEA results for each bulkhead, with the corresponding stress distributions shown in Figures 68, 69, 71, 73, 74 and 76. The principal results from the simulations are summarized in Table 3.

Table 3: Maximum stress and minimum safety factor for each bulkhead

Bulkhead	**Maximum stress (MPa)**	**Safety factor (min)**
BOOSTER - motor	147	1.87
BOOSTER - bottom avionics bay	117.11	2.34
BOOSTER - top avionics bay	40	6.88
SUSTAINER - motor	80	3.45
SUSTAINER - bottom avionics bay	146.6	1.89
SUSTAINER - top avionics bay	88.50	3.11

III.A.3 Interstage

The interstage is primarily composed of three main parts: the booster avionics bay top bulkhead, the interstage tube, and a centering ring. The coupling method used for the interstage relies on the sustainer motor to both align the longitudinal axes of the two stages and transfer axial loads.

Assembly begins with the avionics bay top bulkhead, which is secured to the fuselage using four radial bolts. The interstage tube is then attached to the upper cylindrical section of the bulkhead using four additional radial bolts and positioned by the slot located between the two cylindrical sections of the bulkhead. The sustainer motor is subsequently inserted into the interstage tube until it contacts the bulkhead, where it is axially supported by the cylindrical interface.

The centering ring is attached to the exterior of the interstage tube and serves both structural and assembly functions. It enables rapid stage integration while ensuring alignment between the two stages. Structurally, it also removes bending loads in the sustainer pitch and yaw directions from the motor interface by reacting them instead as radial compression loads. During stage separation, this assembly remains with the booster while the sustainer motor and motor tube slide out of the interstage tube.

Regarding ignition, a standard COTS igniter is used and routed from the top along the exterior of the motor through a tubular channel. It exits at the lower end, where it is compressed between the motor tube and the interstage tube. Because this interface was identified as a potential risk for wire damage, three mitigation measures were implemented to ensure wire integrity and ease of assembly. First, the igniter wiring is divided

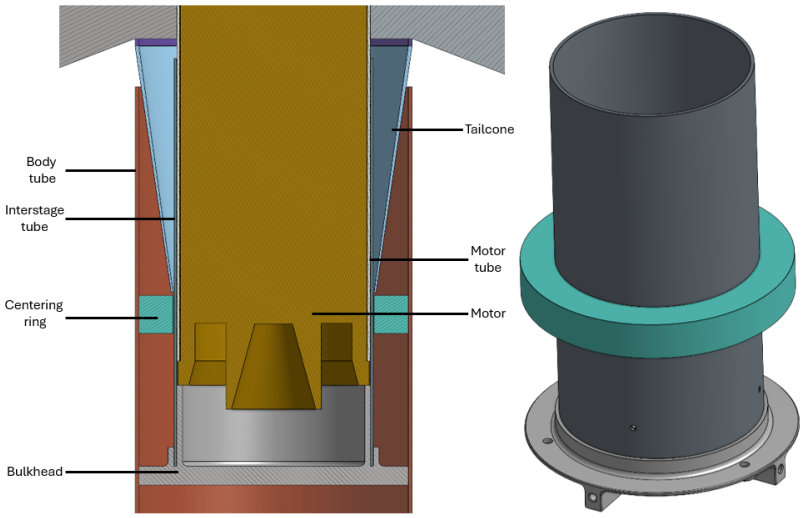


Fig. 8 Interstage system

into sections, with the compressed segment replaced by an ultra-thin flat flex cable suited for the limited clearance. Second, shallow grooves are machined into the motor tube surface to recess the cable below the outer diameter. Third, four independent cables are installed to provide redundant ignition paths in the event of wire failure.

The primary design objective of the interstage is structural integrity under boost loading, with additional emphasis on ease of stage assembly while on the launch pad. As with the overall vehicle, mass optimization remains a secondary objective after structural and operational requirements.

Several alternative interstage concepts were considered, including a conventional tube coupler between the two fuselage sections, which was used successfully on the previous year's rocket and offers high simplicity. However, due to the flight dynamics constraints associated with a 45 kft mission and the resulting need for a sustainer tailcone, a direct body coupler was not compatible with the closed aft geometry.

The main load case considered for this design is thrust transfer from the booster to the sustainer through the bulkhead interface. The anticipated axial load corresponds to the sustainer mass accelerated at 8 g, representing the maximum expected acceleration in the stacked configuration. Analytical calculations based on the beam theory and FEA are performed to verify its resistance and are presented more precisely in the appendix. The resulting stress obtained is 65 MPa and returns a safety factor with Al 6061-T6 of 4.2, adequately over our target of 3.

III.A.4 Integration

Booster Avionics Bay

The booster avionics bay is bounded by two bulkheads 27 and 29. To ensure the structural integrity of the bay, three aluminum threaded rods are symmetrically distributed along its circumference, holding the assembly together. The internal structure of the avionics bay is 3D-printed in four parts, each secured in place by nuts tightened against the rods.

All electronic boards are mounted using thermally inserted M2 and M3 brass inserts, which allow screws to be driven directly into the printed parts for a reliable and repeatable fastening method. The main battery pack is housed in a custom enclosure that secures the electrical connections on both poles of the 21700 cells, with the Battery Management System (BMS) mounted on top.

The arming PCB is positioned at the center of the bay to minimize wire length and improve connection robustness. It features secured enclosures around the arming switches. Each enclosure includes a hole tangent to its switch, allowing a push-pin to be inserted from the exterior of the rocket to arm the system without having

access to the bay.

The cameras are mounted on a sliding module that allows them to be positioned as close as possible to the outer surface of the rocket. This module is fastened to the airframe via screws through the fuselage.

Finally, all antennas are spaced as far apart as possible from each other and from metallic components to minimize radio frequency interference. Long antennas are additionally secured axially to prevent any movement during flight.

The avionics bay design and component annotations are illustrated in Figure 9.

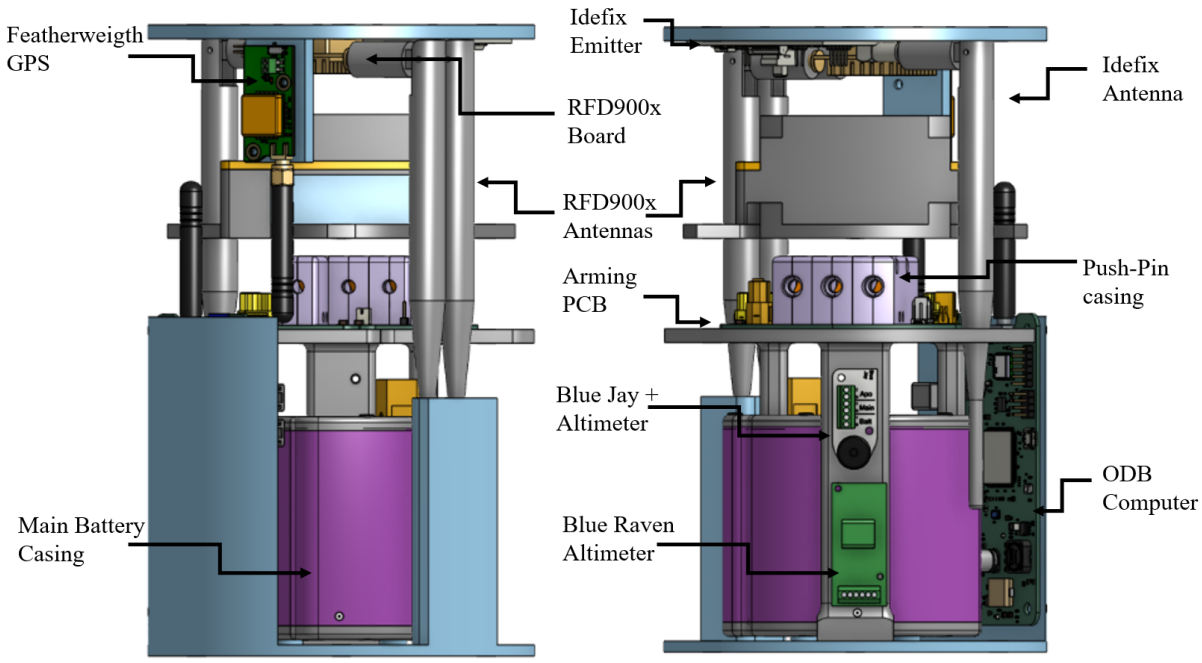


Fig. 9 The left and right images show the rear and front of the booster avionics bay, respectively.

Sustainer Avionics Bay

The sustainer avionics bay is quite similar to that of the booster, bounded by two bulkheads 33 and 35. The main differences are as follows: additional space is allocated above the arming PCB to accommodate the Spot Trace tracker; a dedicated cable channel is reserved at the bottom of the bay to route the motor ignition wire; the camera module is oriented vertically rather than horizontally; the Blue Jay+ is replaced by the Stratologger CF, resulting in different mounting hole configurations; and finally, the bay uses 2 threaded rods instead of 3, placed symmetrically opposite one another.

III.A.5 Fins

The primary design objective of the sustainer fins is to maintain vehicle stability throughout the full flight envelope. Stability margin is evaluated as the distance between the center of gravity (CG) and center of pressure (CP), expressed either in body calibers or as a percentage of total rocket length. While ensuring stable flight, the fins are also optimized to reduce drag and improve aerodynamic performance.

The main design limitation is fin flutter, which imposes minimum thickness and geometric requirements to ensure sufficient structural rigidity. In addition, several competition rules directly constrain the design. Section 10.2 of the DTEG specifies a minimum fin span of 80% of body diameter. Stability requirements also impose a minimum margin of 7.5% in the subsonic regime and 10% in the supersonic regime. Beyond these requirements, the design targets were set at 3 calibers for the full stack at rail exit and 2 calibers for the sustainer alone.

Material selection was the principal trade-off considered, as it strongly influences both structural resistance and aerodynamic performance. The main materials evaluated were aluminum, carbon fiber composites, and higher-strength metals. Aluminum was selected for the sustainer fins due to its reliable mechanical properties, competitive cost, and ease of procurement. The stronger metals considered provided only limited structural benefit for a significant cost increase. Carbon fiber represented an attractive lightweight option, but the lower and less certain mechanical properties would have required a higher safety factor, reducing the expected performance benefit.

Drag separation was also considered in the selection process, as the sustainer should have the lowest practical drag while the booster should produce comparatively higher drag before staging. Aluminum therefore represented the best compromise by enabling thin, reliable, and economical fins. Their thickness is 4.2 mm and the edges are cut sharp with an angle of around 20 degrees.

The booster fins were subject to less demanding constraints, as their impact on total flight performance is lower. In fact, reduced aerodynamic efficiency is beneficial for drag separation, that is why the edges are left square. Stability requirements for the booster fins only apply to the stacked vehicle configuration, with a target margin of 3 calibers. The booster alone after separation was not specifically designed for the same stability level as the ascent phases.

The structural analysis conducted for the sustainer fins focused on flutter resistance. The method used is based on the Apogee Peak of Flight Newsletter 615, which presents a simplified flutter calculation derived from NACA flat-plate theory. The equation uses fin geometry and material properties to estimate the critical flutter velocity. This value is then compared with the maximum expected flight velocity, with a target safety factor of 1.5. For aluminum fins, a shear modulus of 26.2 GPa was used based on published material databases such as MatWeb. The resulting flutter velocity is 726 m/s, compared with an expected maximum velocity of 482 m/s, giving a safety factor of 1.51.

For booster fin flutter resistance, the critical factor is the equivalent structural properties of the composite construction. The fins use a sandwich structure composed of a 4 mm foam core with 1.25 mm carbon fiber skins on each side. The shear and bending properties required for flutter analysis were estimated using classical sandwich theory combined with experimental material data for the composite skins. The resulting equivalent shear modulus used in the calculations was 15.1 GPa. The predicted flutter velocity is 714 m/s, compared with a maximum expected velocity of 310 m/s, resulting in a safety factor of 2.3.

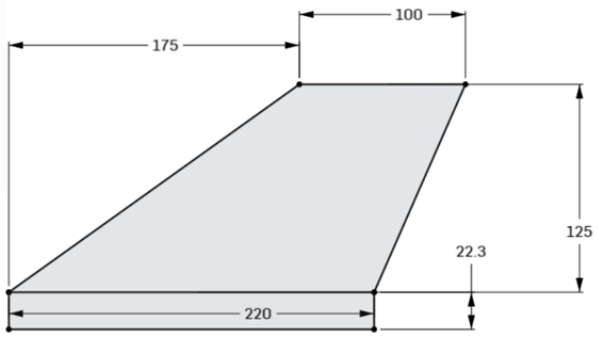


Fig. 10 Sustainer fins dimensions (mm)

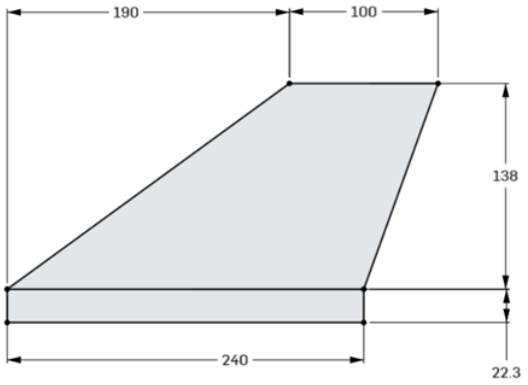


Fig. 11 Booster fins dimensions (mm)

III.A.6 Tailcone

The primary function of the tailcone is to reduce sustainer base drag and thereby improve the achievable apogee altitude. It does so by promoting a more gradual pressure recovery behind the vehicle. By maintaining attached external flow along the converging surface, the wake separates at a smaller cross-sectional area than it would on a blunt aft body. This reduces the low-pressure recirculation region responsible for base drag and therefore decreases overall aerodynamic losses.

The optimal tailcone geometry is a compromise between pressure drag reduction and additional skin-friction drag. A larger cone angle is generally desirable to shorten the component and reduce wetted area, but excessive angle can cause premature flow separation and reduce effectiveness. The selected design therefore seeks the largest practical cone angle that still maintains attached flow, while minimizing overall length and mass.

Structurally, the tailcone is divided into two bonded sections installed inside the lower end of the sustainer body tube. Four slits are incorporated into the internal section to allow clearance and fitment around the fin roots during assembly. This configuration simplifies integration while maintaining continuous external geometry.

The component is manufactured entirely from carbon fiber using a wet layup process over a 3D-printed male mold. Carbon fiber was selected for its compatibility with the rest of the composite airframe and suitability for manufacturing the required conical geometry with available in-house processes. The design was also developed with manufacturability as a key constraint.

The external dimensions of the tailcone were determined through flight performance simulations by optimizing predicted apogee altitude. OpenRocket was used for this trade study due to its rapid iteration capability and ease of geometric modification. Final simulations predict an apogee of 45,500 ft with the tailcone installed, compared with 40,500 ft without it. The resulting gain of approximately 5,000 ft represents a major performance improvement and justifies the added integration complexity associated with the interstage configuration.

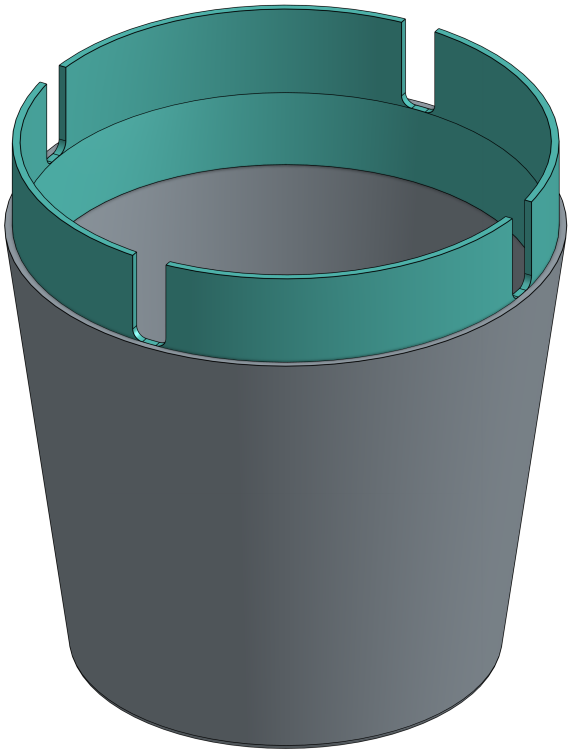


Fig. 12 Sustainer tailcone

III.B Avionics Subsystems

III.B.1 Avionics overview

The avionics systems for our rocket are divided into the booster and the sustainer sections. Each stage is equipped with different avionics systems tailored to their needs.Each subsystem was designed with independent power supplies, redundant deployment mechanisms, and real-time telemetry capabilities to maximize system robustness and flight success. Figure 13 and 14 show block diagrams of the major components of each stage, including arming and power distribution. A total of four independent circuits are used for total redundancy and robustness : one circuit for each deployment altimeter, one circuit for tracking and one for SRAD components. The wiring and circuit diagrams are shown in appendix V.

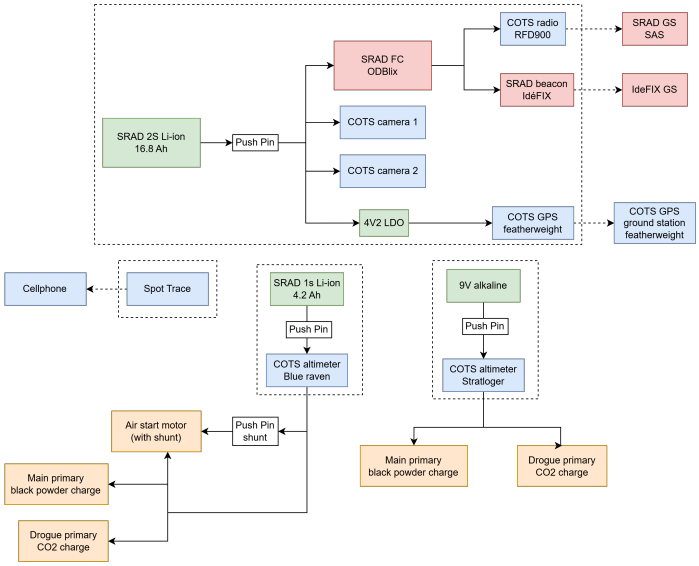


Fig. 13 Block diagram of sustainer avionics subsystem

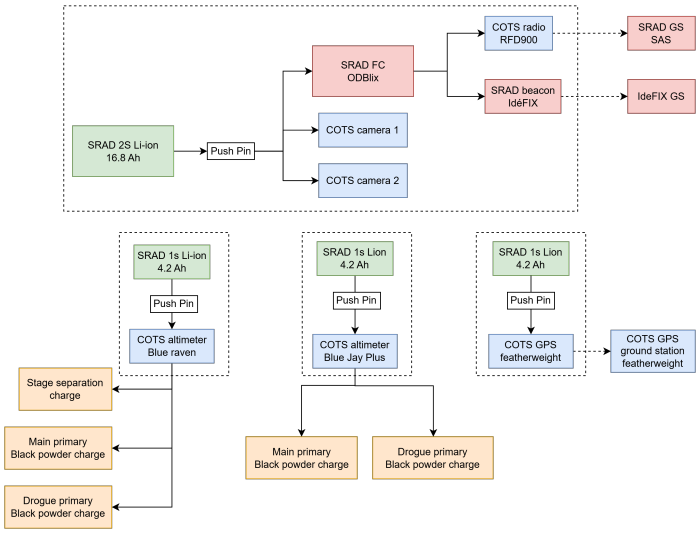


Fig. 14 Block diagram of booster avionics subsystem

The avionics subsystem include vital flight components such as COTS altimeters and trackers. The SRAD portion of this subsystem is used for additional tracking and telemetry. Included in this system is a newly developped SRAD flight computer ”ODBlix”. This flight will also serve as the first flight test of this flight computer, during which a gyroscopic deployment algorithm will be tested.

III.B.2 Deployment altimeters

The following COTS altimeters are used in their respective stage. The settings of these altimeters are detailed in appendix V. All altimeters used are reputable, well known, well documented and flight tested commercial altimeters. The Blue Raven altimeters were chosen for their 3rd deployment channel, allowing for an air-start motor and for the possibility of a stage separation charge in the booster. The impressive amount of flight data recorded by the Blue Raven altimeters as well as their Bluetooth capability were also considered in this decision. The Blue Jay Plus is used in the booster and the Perfectflite Stratologger CF is used in the sustainer, both are used as redundant COTS deployment altimeters.

An important precision to be made is that the Blue Jay Plus' magnetic switch is bypassed to fix our push-pin system.

Stage	Device	Function
Sustainer	Blue Raven Altimeter	COTS main deployment (drogue and main), Air-start motor
	Stratologger CF	COTS redundant deployment (drogue and main)
Booster	Blue Raven Altimeter	COTS main deployment (drogue and main)
	Blue Jay Plus Altimeter	COTS redundant deployment (drogue and main)

Table 4: List of deployment altimeters for each stage

III.B.3 Tracking solutions

Tracking solutions for each stage include at least one COTS GPS tracker and two separate SRAD tracking solutions.

Each stage (deployable payload included) is equipped with a Featherweight GPS Tracker, offering real-time positional tracking throughout the flight. They are a reputable GPS tracker widely used in amateur rocketry. The Featherweight GPS tracker is the official COTS tracker for both stages. The sustainer's Featherweight GPS tracker is powered by the SRAD power system through a 4.2V linear voltage regulator (LDO). This was done because the switch used for the tracker's circuit in the booster in instead used to shunt the motor. We have identified this design choice as a major risk. Indeed all of our tracking solutions are powered by the same power system. This is why we added the Spot Trace in the sustainer, making for an additional and completely independent GPS tracking solution.

Each stage is equipped with an ODBlix Flight Computer, developed in-house. The ODBlix records high-frequency flight parameters including altitude, velocity, GPS coordinates, and system health metrics. During flight, it streams real-time telemetry to the primary ground station via an RFD900x radio modem. Furthermore, it integrates a secondary transmitter to provide a dedicated tracking signal to a specialized ground station named IdeFIX (which can also be used separately).

Each ODBlix unit is interfaced with an RFD900x Radio Modem, enabling long-range communication on the 900 MHz ISM band. The system provides a live data downlink of flight dynamics and GPS positioning through ascent, apogee, and descent. To ensure adaptability at various launch sites, the operating frequency can be reconfigured on-site via a system reboot.

Each stage incorporates an IdeFIX recovery beacon. Operating independently of the primary telemetry stream, the IdeFIX transmits a dedicated radio signal for direction-finding (RDF). The source of the signal can be located using an IdeFIX ground station integrated with a directional Yagi antenna. This beacon offers a recovery option that is independent of GPS. Additionally, this device can also be connected to ODBlix to transmit the real time GPS coordinates, acting as a second radio link for ODBlix. This device becomes particularly usefull during the last 300m of recovery, when GPS the last known GPS coordinates can be quite imprecise.

The Spot Trace is a compact commercial satellite-based GPS tracking device designed for monitoring and recovering assets over long distance. It uses satellite network to transmit position data, enabeling reliable tracking event enven in remote areas without cellular coverage. The device allows users to monitor location, receive alerts, and track movement directly from a smartphone, tablet, or computer through an online mapping

platform. This device is only used for ground recovery and will not be used to track the rocket during flight, since its maximum operating altitude is 21,320ft. It serves as a last resort for recovery, if all other methods fail. This device is compliant with part 15 of the FCC Rules, and operates outside of the student band plan provided by ESRA. Its licensing and legality in the US are guaranteed by the manufacturer. Therefore, it will not interfere with other devices used by the other teams at the IREC.

Stage	Device	Function
Sustainer	Featherweight GPS tracker	Official COTS GPS tracker
	SRAD "ODBlix" Flight Computer	Telemetry and GPS tracking
	SRAD "IdeFIX" Recovery Beacon	Ground recovery
	COTS Spot Trace GPS solution	GPS tracker
Booster	Featherweight GPS tracker	Official COTS GPS tracker
	SRAD "ODBlix" Flight Computer	Telemetry and GPS tracking
	SRAD "IdeFIX" Recovery Beacon	Ground recovery

Table 5: List of tracking solutions for each stage

III.B.4 Telemetry and ground support

The ground station was designed to receive, decode, and visualize telemetry data transmitted from the rocket during flight. It also provided local atmospheric measurements for comparison and redundancy. The system was engineered around a modular and scalable architecture to facilitate integration, robustness, and expandability.

- **SRAD "SAS" Ground Station**

  A Raspberry Pi 4 Model B serves as the primary computing platform. It manages the user interface, receives incoming telemetry via USB-connected RFD900x radio, and communicates with the custom ground station board through a GPIO bus. This computing unit is intefaced with a NEO-8M GPS Module, a BMP280 Barometric Sensor for local pressure and altitude readings, two LCD displays, a touchscreen and costum buttons. It is powered by a 5V battery pack and cooled via fans. This costum base-station is all integrated into a portable rugged case.

- **Telemetry Reception**

  The RFD900x radio modem received live telemetry data streamed from the ODBlix flight computers onboard the rocket. Data packets included are same as Telemetry Protocol and Data Structure A custom software interface on the Raspberry Pi processes and displays the incoming data in real time on 2 LCDs and a touch screen.

- **IdeFIX substation** The IdeFIX substations is a portable device designed to locate the rocket based on transmitting power on the 433 MHz band. It will assist the recovery team during the final steps of the recovery or when the team is within a 200m range from the rocket.

III.B.5 Arming and power

Booster
To minimize wiring between components, batteries, and switches, an arming PCB consolidates all switches onto a single board and serves as a central connection hub for all components. Although it is a single PCB, all circuits are completely independent and electrically isolated from one another. The complete wiring diagram of the booster avionics bay is presented in appendix V.

The main battery connects to the arming PCB via an XT60 connector. It is a 2S Li-ion powering the Idefix transmitter, the ODB, the cameras, and the RFD-900x, all controlled by a single shared switch. The Blue Raven, Blue Jay+, and Featherweight each have their own dedicated 1S Li-ion battery and independent switch.

All battery connections include a reverse polarity protection circuit, which cuts power if a battery is inserted in the wrong orientation. Additionally, each independent circuit uses a unique connector type, making incorrect cross-connections physically impossible.

A housing screwed onto the PCB encloses each switch mechanism, constraining the actuating screw and preventing accidental disconnection during flight. A nut is also used to make sure the screw does not fall from the casing.

Regarding the altimeter power supply, each altimeter is powered by a dedicated 1S Li-ion battery with a capacity of 4200 mAh. The Blue Raven has an average current draw of 20 mA, yielding a battery life of 210 hours. The Blue Jay+ draws an average of 16 mA, resulting in a battery life of 262.5 hours. Finally, the Featherweight altimeter has a maximum current draw of 112 mA, giving a worst-case battery life of 37.5 hours. In all three cases, the battery capacity is largely sufficient for the mission.

Regarding the main system power supply, the two cameras together draw 600 mA, the Idefix transmitter draws an average of 220 mA, and the ODB combined with the RFD-900x draw an maximum of 900 mA, for a total current consumption of 1720 mA. With the main 2S battery capacity of 16.8 Ah, this allows for a total operating time of 9.8 hours, which is sufficient.

Sustainer

Similarly to the booster, the sustainer features an arming PCB with a comparable design, where each circuit has its own dedicated switch and power supply. The complete wiring diagram of the sustainer avionics bay is presented in Figure 82.

The first switch powers the main system, so the the Idefix transmitter, the ODB, the cameras, the RFD-900x, and the Featherweight altimeter via a 2S Li-ion battery of 16.8 Ah. Since the Featherweight is now part of this shared circuit, the total average current draw increases to 1832 mA, yielding an operating time of 9.2 hours.

The second switch powers the Blue Raven on a dedicated 1S LiPo battery of 4200 mAh. With an average current draw of 20 mA, this provides a battery life of 210 hours. The third switch powers the Stratologger CF via a 9V alkaline battery rated at 450 mAh. At an average draw of 1.5 mA, this yields a battery life of 300 hours. Finally, the fourth switch is used to shunt the sustainer motor igniter, adding an additional layer of safety by ensuring the motor cannot be inadvertently ignited.

Additionally, the sustainer carries a Spot Trace GPS tracker operating on its own independent battery. Combined with the onboard altimeters, this provides full redundancy for both altitude measurement and GPS tracking systems.

III.B.6 Communications

Ground communications are a vital part of this mission. Ozyman uses various RF transmitters in different bands to communicate with our ground support equipment. Table 6 shows the complete list of all RF transmitters to be used in all stages of Ozyman, as well as their mode of operation, frequency band, and output power.

Stage	Transmitter	Operating freqs	Mode	Bandwith	Power	Note
Sustainer	RFD900x	902 - 909 MHz	B or A	140 kHz or 650 kHz	1W	Frequency allocation as early as possible would be appreciated, as modification is inconvenient
	Featherweight GPS tracker	910 - 928 MHz	A	800 kHz	100 mW	
	IdeFIX recovery beacon	420.6 - 438 MHz	C	100 kHz	100 mW	
	Spot Trace	1611.25 - 1618.75 Mhz	-	-	300 mW	
Booster	RFD900x	902 - 909 MHz	B or A	140 kHz or 650 kHz	1W	Frequency allocation as early as possible would be appreciated, as modification is inconvenient
	Featherweight GPS tracker	910 - 928 MHz	A	800 kHz	100 mW	
	IdeFIX recovery beacon	420.6 - 438 MHz	C	100 kHz	100 mW	
Payload	Featherweight GPS tracker	910 - 928 MHz	A	800 kHz	100 mW	
	IdeFIX recovery beacon	420.6 - 438 MHz	C	100 kHz	100 mW	

Table 6: List of all RF transmitters used in Ozyman

III.C Propulsion Subsystems

III.C.1 Thrust Structures

Booster

Requirements The booster thrust plate transmit the motor thrust to the airframe during the boost phase and retain the motor at parachute deployment, while accommodating the surrounding components (Aeropack retainer, lower centering ring, fins, rail button and motor tube) and minimizing mass.

Design The thrust plate (fig. 23) is machined from 6061-T6 aluminium, pressed against the lower end of the booster body tube and secured with four M5 screws and epoxy. The motor tube is bonded at its center, allowing it to also act as a centering ring. A two-part COTS Aeropack retainer is screwed to the underside, clamping the motor in both axial directions. A forward-retaining alternative was considered for its mass advantage but rejected, as incomplete threading of the motor retaining rings would shift the assembly position and complicate integration.

Analysis The thrust plate is analyzed under two load cases: the motor thrust during the boost phase and the empty motor weight reacted at parachute deployment (conservatively taken as 20 G). Shear and bending stresses were evaluated analytically using Roark's formulas for circular flat plates with fixed outer edges, and validated by FEA in Siemens NX. The detailed calculations are presented in Section V of the appendix (figure 67). Bending is the dimensioning mode, with a minimum safety factor of 2.89 against the 6061-T6 yield strength.

Sustainer

Requirements The sustainer motor bulkhead transmit the booster-to-sustainer thrust during the boost phase, transmit the sustainer motor thrust during its burn, anchor the main parachute at recovery deployment and retain the motor afterwards. It must also accommodate the motor assembly length variations caused by incomplete threading of the retaining rings, while minimizing mass.

Design The sustainer motor bulkhead (fig. 31) is also machined from 6061-T6 aluminium and fixed to the fuselage with four radial M5 screws. A central eye-bolt is secured with a nut, with its threaded section protruding underneath to screw directly into the motor forward closure, thus serving simultaneously as the parachute anchor point and the motor connector. This integrated design accommodates motor length variations by adjusting the eye-bolt height with washers, ensuring the aft end of the motor remains at a constant position relative to the top, which is critical for the interstage assembly. A separate-function alternative (dedicated thrust plate, retainer and parachute bulkhead) was considered but rejected, as adding a sustainer tailcone and components at the motor aft end would significantly complicate the interstage integration and increase mass.

Analysis The bulkhead is analyzed under three load cases corresponding to its three functions: the sustainer motor thrust (2100 N from the thrust curve), the booster-to-sustainer thrust transmission computed from the sustainer mass at the maximum stack acceleration of 8 G (2275 N), and the parachute deployment load conservatively estimated from a fully-inflated main canopy at drogue descent speed (1540 N on the bulkhead side). The booster thrust transmission is the dimensioning case. Bending stresses were first evaluated analytically using beam theory on the central transversal ribs, and validated by FEA in Siemens NX. The detailed calculations and FEA results are presented in Section V of the appendix (figure 68), with a minimum safety factor of 3.45 against the 6061-T6 yield strength.

III.C.2 Motors

For this two-stage rocket, both motors use COTS solid propellant systems manufactured by Aerotech. The first stage is powered by an N4000 motor, which provides the initial propulsion for the full stack, while the second stage uses an N1000 motor as the sustainer. The main characteristics of both motors are summarized in Table 7, followed by their respective thrust curves obtained from thrustcurve.org. The combined total impulse of the vehicle, corresponding to the sum of both motors, is 32,310 Ns.

Placement	Designation	Dimensions [mm]	Loaded weight [kg]	Total impulse [Ns]	Burn time [s]	Maximum thrust [N]	Propellant
Booster	N4000	98 x 1433	16.4	18 184	4.7	4750	White Lightning
Sustainer	N1000	98 x 1046	12.8	14 126	13.1	2120	White Lightning

Table 7: Motor specifications

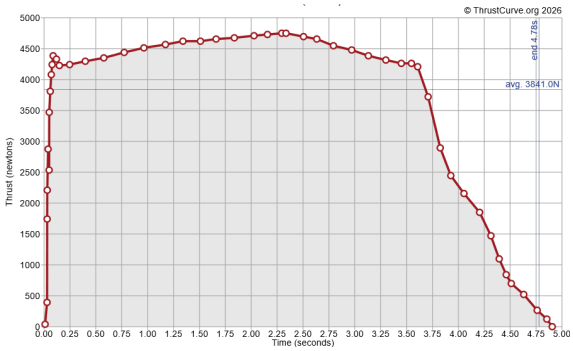


Fig. 15 N4000 thrust curve

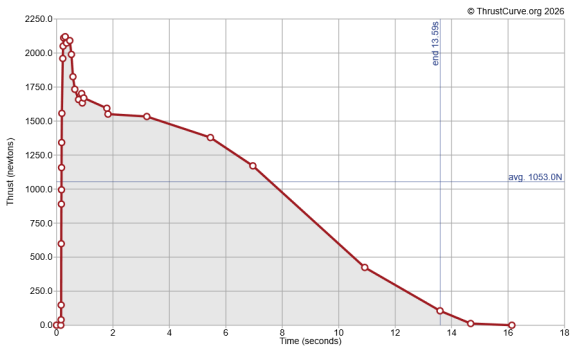


Fig. 16 N1000 thrust curve

The selection of these motors was primarily driven by availability, as both are manufactured by Aerotech and distributed by Wildman, the official IREC vendor. From a performance standpoint, two specific propulsion requirements were identified for a multistage rocket targeting high altitude.

First, the booster motor must provide high lift-off thrust to generate sufficient initial acceleration and rail-exit velocity for a rocket of this size and mass. The design requirement was to maintain a thrust-to-weight ratio greater than 8 at launch.

Second, the sustainer motor should provide a relatively long burn duration to reduce drag losses for a given total impulse. Since aerodynamic drag scales with the square of velocity, it is generally more efficient to sustain a lower velocity over a longer period than to reach a short-duration high-speed peak.

III.D Recovery Subsystems

III.D.1 Deployment Systems

Requirement	**Description**
Electronic triggering	System must be electronically activated
Redundancy	Maximize redundancy to minimize failure risk
Gentle actuation	Deployment must avoid damaging the payload
Environmental tolerance	Withstand acceleration, pressure, temperature, and vibration
Ignition reliability	Reliable ignition mechanism for deployment sequence

Design The design is divided into three independent deployment systems: one utilizing high-pressure carbon dioxide (CO_2), and the others based on black powder charges. One is simply a black powder charge well, it is used for the deployement of the main parachute of the sustainer. It is an easy design, it's reliable and capable of pressurizing a big compartement like the tube containing the main parachute and the payload. The other system is the line cutter system. It acts a bit like a Jolly Logic Chute Release but is directly connected to the avionics bay and breaks open a tie wrap using black powder and a hammer.

We begin with the CO_2 system. It employs a 25-gram high-pressure CO_2 cartridge, typically used for tire inflators. The cartridge is threaded and screwed onto a custom base mounted to the rocket's bulkhead. Integrated

into this base is a small black powder chamber containing an electric igniter. Positioned above the black powder charge is a spring-loaded spike with an o-ring to seal the black powder, carefully aligned with the weakened section of the cartridge. The spring ensures that the spike remains retracted and cannot pierce the cartridge prematurely. Upon ignition, the black powder generates a rapid pressure increase that drives the spike into the cartridge, rupturing the seal and releasing the compressed CO_2 to actuate the deployment mechanism.

The black powder system features a base with a design similar to that of the CO_2 system, but instead of housing a cartridge, it is filled entirely with black powder. An electric igniter is embedded within the powder to initiate the deployment sequence. The chamber is securely sealed on top to prevent any leakage of the black powder, ensuring both safety and reliable ignition. Upon activation, the rapid combustion of the black powder generates a high-pressure burst that drives the mechanical deployment.

The final system, the line cutter is held around the parachute with tie wraps to keep it closed. An igniter is inserted in the aluminum case with a very small amount of black powder. At the desired altitude, the e-match lights the black powder to break open the tie wrap and release the parachutes. For redundancy, each deployment stage is equipped with two identical systems of the same type. The second system is triggered a few seconds after the first, serving as a backup in case of failure and ensuring reliable deployment.

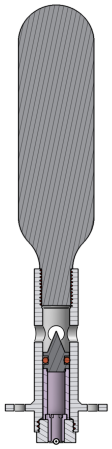


Fig. 17 Cross section of the assembly of the SRAD CO2 deployment system

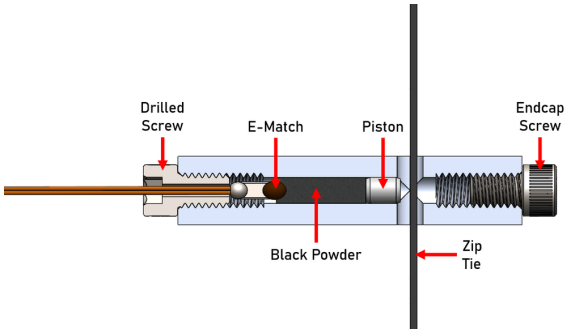


Fig. 18 Cross section of the assembly of the line cutter system

III.D.2 Parachutes and Harness

Sustainer Deployment Architecture: Dual-deploy configuration with a drogue bay in the nosecone and a main bay between the avionics and motor, enabling simultaneous main parachute and payload deployment at low altitude. The exact recovery schematics for the sustainer with the length of the shock cord and positioning can be found in figure 79 in appendix V.

Sustainer Parachutes: The 84 in toroidal ($C_D = 2.2$, Rocketman) was selected as main, yielding 6.51 m/s nominally and 7.84 m/s in the aborted-ignition contingency, both within the $<$11 m/s requirement. The 72 in was rejected due to higher landing speed and non-ownership; the 96 in was rejected on mass and packed-size grounds. For the drogue, the 24 in elliptical ($C_D = 1.55$, Fruity Chutes) was selected: it is the only size satisfying the 20–40 m/s requirement under all scenarios (18 in falls outside range at aborted ignition), and the higher descent speed of the elliptical variant reduces wind drift compared to the toroidal alternative.

Booster Deployment Architecture: The booster uses a single parachute bay with a line-cutter system. Both the drogue and the main parachute exit the rocket simultaneously at apogee on the same shock cord; the main parachute remains closed, restrained by tie-wraps, until the deployment system severs them at the target altitude. The operation of this line-cutter system is detailed in the following sections. The exact recovery schematics for the booster with the length of the shock cord and positioning can be found in figure 80 in appendix V.

Booster Parachutes: All candidates satisfy the <11 m/s main requirement (Table 8). Selection was narrowed to team-owned hardware: the 84 in elliptical ($C_D = 1.55$, 7.19 m/s, Fruity Chutes) was chosen over the 96 in toroidal for its lower mass and packed size, critical given both parachutes share a single bay. For the drogue, the 18 in elliptical ($C_D = 1.55$, 35.48 m/s, Fruity Chutes) was selected over the 24 in for its smaller packed size, lower mass, and reduced wind drift; deployment forces remain within harness-rated limits as confirmed in Table 9 and it's following analysis.

Analysis

In the analysis, we check the descent speed of both stages using simulation. For the drogue, we simulate the terminal velocity with parachutes sizes of 18, 24 and 36 inches for 2 different geometry variants giving a CD of 1.55 or 2.2. For the main, we simulate for diameters of 72, 84 or 96 inches with the same options of 1.55 or 2.2 drag coefficients. The simulations were done on Open Rocket and on our SRAD simulator called SimulOutil. They featured the rocket geometry, weight, launch environment of the IREC launch pad and without winds. The results are shown in the following tables.

Drogue

Dia. (in)	Geometry	C_D	v_{booster} (m/s)	$v_{\text{sustainer}}$ (m/s)
18	Elliptical	1.55	35.48	38.78
18	Toroidal	2.20	34.09	37.41
24	Elliptical	1.55	26.70	29.15
24	Toroidal	2.20	22.07	23.14
36	Elliptical	1.55	17.96	19.57
36	Toroidal	2.20	13.73	15.11

Main

Dia. (in)	Geometry	C_D	v_{booster} (m/s)	$v_{\text{sustainer}}$ (m/s)
72	Elliptical	1.55	8.43	9.21
72	Toroidal	2.20	7.06	7.72
84	Elliptical	1.55	7.19	7.85
84	Toroidal	2.20	5.95	6.51
96	Elliptical	1.55	6.19	6.74
96	Toroidal	2.20	5.08	5.55

Table 8: Descent speed analysis for different parachute options (booster and sustainer stages).

The load distribution at each bulkhead is calculated using the derived equation (16) for a load distribution in a two-anchor rope system shown in Appendix V (see page 65). We can now use this result to calculate the applied force on a bulkhead due to the opening of a parachute. In each stage we design for worst possible scenario.

	Drogue apogee	Drogue late apogee	Main nominal	Main at apogee	Main late apogee
Sustainer	97.39 lbf	103.29 lbf	895.79 lbf	1750.70 lbf	1803.66 lbf
Booster	26.84 lbf	35.48 lbf	946.94 lbf	545.36 lbf	777.61 lbf

Table 9: Forces acting on recovery harness due to parachute opening for different recovery event possibilities. A typical multi level wind profile from launch site was used with a ground level wind near max allowable speed for launching to get better cross-range velocity prediction

III.E Payload Subsystems

The 2026 payload is divided into two concepts. A deployable mini-rover and a hyperspectral camera.

III.E.1 Mechanical Design

The rover's mechanical design is optimized to be functional on rough terrain and in significant heat. The rover is printed mostly in PLA in light colors to reduce heating. The parachute is attached to a cable cutter, which is secured to the rover. Even if the parachute is mounted on one side, the rover remains fully functional, even if it lands on the opposite side. Two 6V micro metal gearmotors are mounted on the front wheels, while the rear wheels remain passive. Two different wheel sets were designed for the front and rear. The front wheels are textured to increase grip and improve traction, while the rear wheels are smoother to reduce friction and allow easier movement while rotating. The wheel's design is a honeycomb pattern printed in TPU. This geometry provides good shock absorption while remaining lightweight.

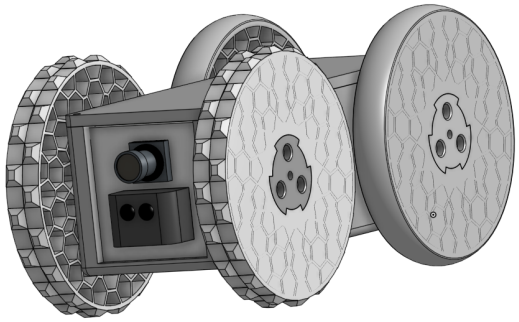


Fig. 19 Rover mechanical design overview

The CubeSat is a 3U format. A shock cord passes through the structure to assist the rover's ejection. The hyperspectral camera is rigidly mounted on the upper right section of the CubeSat and is not part of the deployed system. Weights are placed on the upper left section to maintain mass symmetry. The CubeSat is printed in onyx and reinforced with carbon fiber in regions of high stress concentration. A simulation on the CubeSat's structure has been conducted on Onshape for an onyx CubeSat. The CubeSat drawings are illustrated in annexx** and demonstrates the maximum deformation caused by 1000 Newton which is equivalent to the maximum force applied on the CubeSat during its ejection.

III.E.2 Electronics

The rover electrical system is built around a STM32F429VIT6 microcontroller coordinating all subsystems via I²C, SPI, and UART. The battery pack feeds two buck converters providing regulated 3.3 V and 5 V rails to all onboard components. Inertial sensing is handled by a BNO055 IMU over I²C, while the MS580301BA01-00-TE altimeters provide redundant pressure and altitude readings. The sensor data dictates the rover's behaviour: depending on the rocket's state, the rover operates in different modes. During flight, it remains in standby with movement disabled; motion only activates once landing is detected. Once on the ground, a TF-Luna LiDAR provides real-time short-range obstacle detection, two FIT0487 motors drive the rover's movement, and a RunCam Split 4 V2 captures onboard footage independently. A MEM2067 SD card module logs all sensor data and flight events throughout the mission. An HM-11 Bluetooth module is also included to provide access to debug information during ground operations. The block diagram of the rover electronics is shown in the following figure:

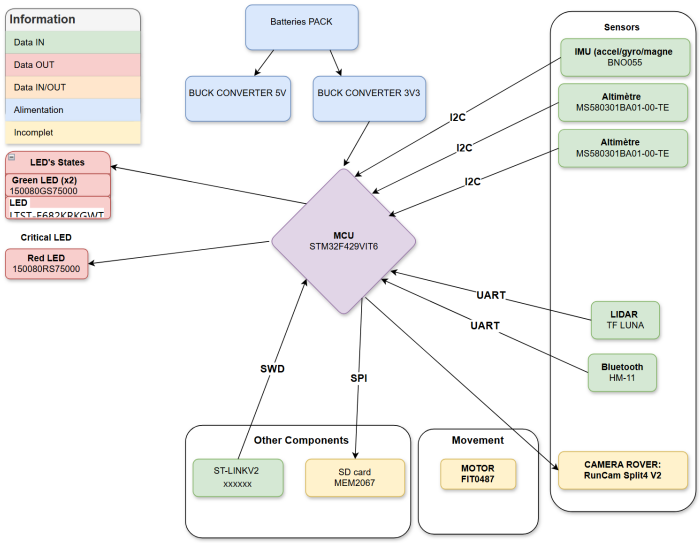


Fig. 20 Block diagram of Rover subsystems

III.E.3 Recovery Systems

The rover will be deployed at 1500ft with the sustainer's main chute and will be ejected naturally when the parachute is fully deployed. The shock cord will be permanently attached to the CubeSat, carrying it with itself when the charges cause the separation at 1500 ft. The rest of the cord will pass through the CubeSat to the motor's bulkhead. When the CubeSat is fully cleared of the rocket's main tube, the shock cord will be fully in tension causing the rover to fall out of the CubeSat. The rover will use a 30-inch elliptical Fruity Chute, providing a nominal descent rate of approximately 6 m/s. A line cutter (as shown in Figure 18) fixed to the rover will allow separation from its parachute on the ground. The cable cutter has successfully been tested in pass rockets. A 200g mass will be added to the rover's parachute to make sure it won't drift away after being cut off. This design has been tested and validated. Both the rover and its parachute will be recovered using a Featherweight GPS Tracker secured inside the rover. There is also an IdeFIX recovery beacon.

III.E.4 Atmospheric Analysis Division

The payload also includes a fixed optical system, whose main objective is to gather atmospheric spectral data throughout the flight. The system is composed of the following components: SSD, Raspberry Pi, Gyroscope, and hyperspectral Camera.

Hyperspectral Camera : The main component of the system is the hyperspectral camera. The entire optical system and associated electronics are provided through a loan agreement, courtesy of Spectron Optics (a Canadian startup founded by Université Laval alumni). Although the specific design of the camera is protected under an intellectual property agreement, a general overview is provided below:

- The hyperspectral technique used is push-broom, in which the sensor retrieves spectral data along a lateral spatial line. The stored data is thus a timestamped spectral $\times$ spatial grid.
- The entire optical system is designed and optimized to fit within a compact $60 \times 40 \times 90$ mm frame.
- The camera has a 17.8 mm aperture, an f-number of $F/2.4$, and a 16 FOV.
- The spectral range spans from 400 nm to 1000 nm with 620 spectral bands. The refractive element is produced in collaboration with the Center for Optics, Photonics and Lasers (COPL).
- The sensor is capable of acquiring 1600 fps at an 8-bit resolution, and a minimum of 940 fps at a 12-bit resolution.
- The data is stored with a timestamp, and the image format can be set via a command through the USB-C 3.1 port.

Main logic : Data flow and writing are managed by a Raspberry Pi (), which serves as the brain of the system. It accepts data from the accelerometer and the camera, synchronizes the timestamps, and saves the files to the SSD.

Gyroscope : To calculate chemical concentrations using Rayleigh scattering, each spectrum must be associated with a spherical orientation vector. To acquire this vector continuously, a high-speed gyroscope, LSM6DSOX, is used. It is connected to the main board via a Teensy microcontroller module.

Storage : Given the high-speed flux of heavy image files, the data storage module must meet stringent performance requirements. Therefore, the system is equipped with an SSD drive (512 GB).

IV Mission Concept of Operations Overview

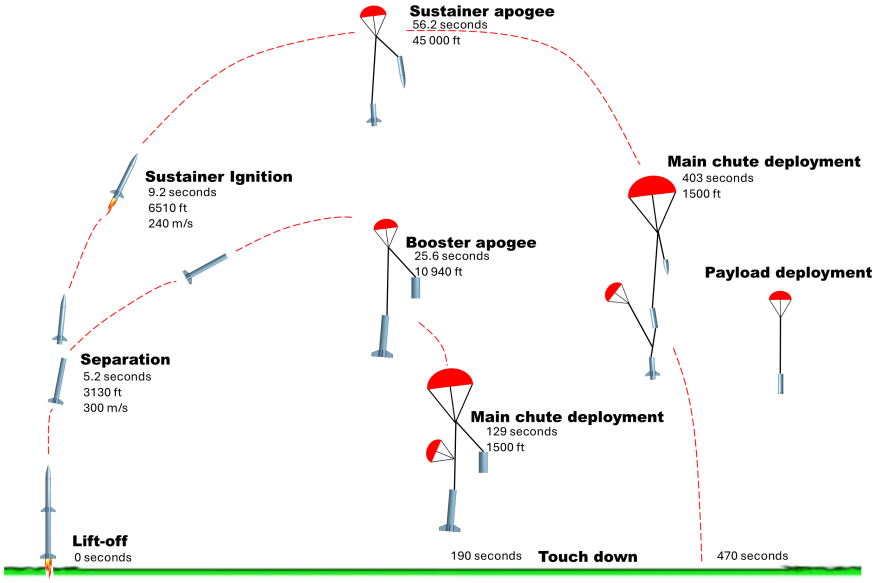


Fig. 21 CONOPS

Ground Handling
Start: The rocket has left the main assembly area and is in transit to the launchpad. The push pins are out of the rocket.
End: The launch rail has been raised and secured and the rocket is ready for launch.

Once all components are assembled and the rocket is transported to the launchpad, the sustainer igniter is prepared and tested with power outside of the motor. The rocket is then mounted on the rail, and the avionics system is powered on by inserting push pins one at a time, following the predefined order outlined in the checklist.

Booster Ignition
Start: The ignition signal has been sent through the igniter.
End: The rocket begins to accelerate.

Within the propulsion subsystem, the propellant will ignite resulting in the rocket accelerating up the launch rail.

Liftoff
Start: The rocket has begun to accelerate.
End: Only one rail button is in contact with the rail.

The thrust produced by the booster motor lifts the rocket off the ground and accelerates it to 29 m/s before leaving the rail.

Powered Ascent
Start: The rocket has left the rail
End: Burnout of the first motor

The rocket ascends freely, under the thrust of the motor and stabilized only by the fins. The motor burns out in 4.76 seconds and the maximum thrust encountered is 4750 N.

Stage Separation

Start: Burnout of the first motor
End: Lost of contact between the stages

When the thrust of the motor ends, the rocket is only subjected to the aerodynamic loads and the sustainer slides out of the interstage in 0.15 second. The pyrotechnic backup charge fires one second after motor burnout.

First Coast Phase
Start: Lost of contact between the stages
End: Ignition of sustainer motor and Apogee of booster

After being separated, both stages continue their ascent completely freely. The sustainer fins keep it's trajectory straight and the booster is highly slowed with the forward end now open.

Sustainer Ignition
Start: The ignition signal is sent through the igniter
End: The sustainer begins to accelerate

The sustainer motor is ignited with the featherweight blue raven altimeter firing the electric signal through the igniter 2.6 seconds after burnout of the booster. The thrust thus begins four seconds after burnout of the booster motor.

Abort Conditions: For the ignition signal to happen, the altimeter must respect the condition of trajectory orientation measured as a minimum altitude planned at that precise moment.

Sustainer Powered Ascent
Start: The sustainer motor produces thrust
End: Burnout of sustainer motor

The upper stage is accelerated again during 13.4 seconds and endures a max thrust of 2120 N.

Second Coast Phase
Start: Burnout of sustainer motor
End: Apogee of sustainer

After the burnout of the sustainer motor, the rocket continues its ascent stabilized by the fins and decelerated by drag forces and gravity.

Booster Apogee and Parachute Deployment
Start: The booster has reached its apogee
End: The Parachute has deployed and inflated

When the air pressure stops decreasing, meaning the apogee of 10 940 ft has been reached, the first altimeter fires a pyrotechnic charge, separating the rocket in two parts and freeing the parachute and recovery harness. The cords extend and the parachute begins inflating. The redundant pyrotechnic charge controlled by the second altimeter fires one second after the first.

Booster Descent Under Drogue Parachute
Start: Cords and parachute are fully extended
End: Ignition of main parachute deployment system

The drogue parachute of the booster slows it down to a velocity of 30 to 27 m/s during the descent until it reaches 1500 ft.

Sustainer Apogee and Drogue Deployment
Start: The sustainer has reached apogee
End: The drogue parachute has deployed and inflated

When the air pressure stops decreasing, meaning the apogee of 45 000 ft has been reached, the first altimeter fires a CO2 charge, separating the rocket in two parts and freeing the parachute and recovery harness. The cords extend and the parachute begins inflating. The redundant CO2 charge controlled by the second altimeter fires one second after the first.

Sustainer Descent Under Drogue Parachute
Start: Cords and parachute are fully extended
End: Ignition of main parachute deployment system

The drogue parachute of the sustainer slows down the rocket from a velocity of 55 m/s to 29 m/s during the descent until it reaches 1500 ft.

Booster Main Parachute Deployment
Start: Ignition of deployment system
End: The main parachute has fully deployed and inflated

When an altitude of 1500 ft is detected, the first altimeter fires a pyrotechnic charge, separating the rocket in two parts and freeing the parachute and recovery harness. The cords extend and the parachute begins inflating. The redundant pyrotechnic charge controlled by the second altimeter fires at 1300 ft.

Final Descent Under Main
Start: Cords and parachute are fully extended
End: The first part of the rocket touches the ground

The main parachute of the booster slows down the rocket to a velocity of 7.2 m/s during the descent until touchdown 190 seconds after liftoff.

Sustainer Main Parachute Deployment
Start: Ignition of deployment system
End: The main parachute has fully deployed and inflated

When an altitude of 1500 ft is detected, the first altimeter fires a pyrotechnic charge, separating the rocket in two parts and freeing the parachute and recovery harness. The cords extend and the parachute begins inflating. The redundant pyrotechnic charge controlled by the second altimeter fires at 1300 ft.

Final Descent Under Main
Start: Cords and parachute are fully extended
End: The first part of the rocket touches the ground

The main parachute of the booster slows down the rocket to a velocity of 6.6 m/s during the descent until touchdown 470 seconds after liftoff.

Touchdown
Start: The first part of both stages has touched the ground
End: The rocket and parachutes have stopped moving

The booster and sustainer land completely and slowly on the ground, waiting to be recovered

Recovery Operations
Start: Landing of the rocket
End: Recovery team has returned to mission control

The GPS positions of both stages are determined, as well as the planing of a route to recover them. After arriving at the rocket's landing spot, the avionics is disarmed and deployment charges are verified. The rocket is then brought back to mission control for final inspection.

V Conclusion and Lessons Learned

The Ozyman project represents a significant milestone for GAUL in the development of complex two-stage high-power rockets. Through the integration of advanced aerostructural design, redundant avionics systems, and multi-method flight simulations, the team successfully developed a vehicle capable of targeting a 45,000 ft apogee while carrying a complex payload.

One of the key strengths of this project lies in the extensive use of simulation and cross-validation. The combined use of OpenRocket, RasAero, RocketPy, and the internally developed SimulOutil improved confidence in performance predictions and highlighted the importance of understanding the limitations of each modeling approach. This redundancy proved essential in identifying discrepancies and guiding design decisions.

From a technical perspective, the project emphasized the importance of balancing performance, reliability, and manufacturability. The use of composite airframes combined with aluminum structural components enabled efficient mass optimization while maintaining acceptable safety factors. Similarly, the avionics architecture demonstrated the necessity of redundancy, with multiple independent deployment, tracking, and telemetry systems significantly reducing the risk of mission failure.

However, several important lessons were identified. First, the absence of full-scale flight testing prior to competition introduces uncertainties that cannot be fully mitigated through simulation alone. Future projects would benefit from incremental validation through subsystem or subscale testing. Second, system complexity particularly in interstage design and recovery mechanisms—must be carefully managed, as it directly impacts integration time, reliability, and operational risk.

From a team management perspective, this project reinforced the importance of clear communication and structured organization. Strong coordination between subsystem leads was essential to ensure system-level consistency. Regular design reviews and cross-team discussions proved critical in identifying integration issues early. Additionally, maintaining clear and accessible documentation throughout the project significantly improved efficiency and knowledge retention.

Finally, knowledge transfer remains a key factor in long-term team success. Establishing structured documentation practices, standardized design procedures, and mentoring of newer members ensures continuity and allows future teams to build upon previous experience rather than restarting from first principles.

In conclusion, the Ozyman project demonstrates GAUL's ability to design, integrate, and manage a complex rocket system. The lessons learned throughout this project will directly contribute to future developments, particularly in increasing system reliability, improving validation strategies, and enabling more ambitious missions.

Appendix A - System Weights, Measures and Performance Data

Table 10: Basic Rocket Vehicle Information

Rocket Parameters	Stack	Booster	Sustainer
Number of stages	2	-	-
Length (mm)	5390	2485	3035
Airframe diameter (mm)	142	142	142
Number of fins	-	4	4
Fin semi-span (mm)	-	138	125
Fin tip chord (mm)	-	100	100
Fin root chord (mm)	-	240	220
Fin thickness (mm)	-	6.5	4.2
Weight on launch rail (kg)	52.9	26	26.9
Propellant weight (kg)	17.78	9.83	7.95
Empty motor case/structure weight	11.42	6.57	4.85
Payload weight (kg)	2	-	2
Center of pressure from nosecone (mm)	3490	-	2222
Center of gravity from nosecone (mm)	3010	-	1916
Effective launch rail length	5639 mm (18.5 ft)	-	-

Table 11: Propulsion Information

Rocket Parameters	Stack	Booster	Sustainer
Motor type	-	Solid	Solid
COTS/SRAD	-	COTS	COTS
Manufacturer and designation	-	Aerotech N4000	Aerotech N1000
Motor letter classification	O	N	N
Average thrust (N)	-	3837	1047
Total impulse (Ns)	32310	18184	14126
Burn time (s)	-	4.7	13.1

Table 12: Predicted Flight Data

Rocket Parameters	Stack	Booster	Sustainer
Liftoff thrust to weight ratio	8.8	-	-
Rail departure velocity (m/s)	29.2	-	-
Minimum static margin	2.4	-	2
Maximum acceleration (G)	8.84	8.84	6.53
Maximum velocity (m/s)	-	305	494
Fin flutter velocity (m/s)	-	714	726
Target apogee (ft)	-	-	45000
Predicted apogee (ft)	-	10950	45000

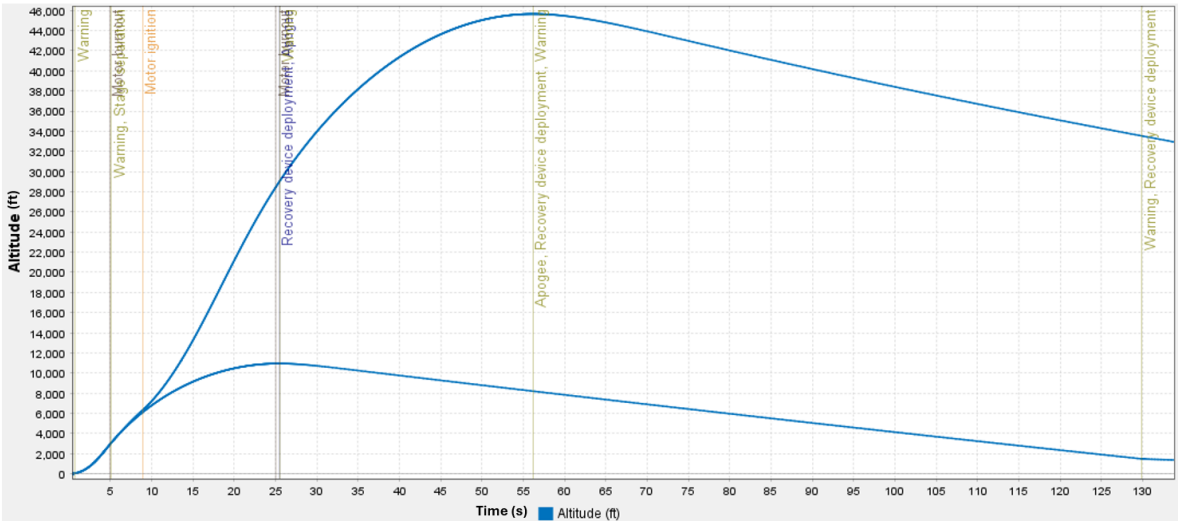


Fig. 22 Flight profile graph

Table 13: Recovery information

Recovery information	Values
Booster Drogue	
Primary altimeter	Blue Raven
Redundant altimeter	Blue Jay Plus
Primary deployment charge	3.5g Black powder
Redundant deployment charge	4g Black powder
Deployment altitude	10900 ft
Descent rate	26.68 m/s
Booster Main	
Primary altimeter	Blue Raven
Redundant altimeter	Blue Jay Plus
Drogue Primary deployment charge	Line cutter
Drogue Redundant deployment charge	Line cutter
Deployment altitude	1500 ft
Descent rate	7.18 m/s
Sustainer drogue	
Primary altimeter	Blue Raven
Redundant altimeter	StratoLoggerCF
Primary deployment CO2 charge	25g
Redundant deployment CO2 charge	25g
Deployment altitude	45 000 ft
Descent rate	28.83 m/s
Sustainer main	
Primary altimeter	Blue Raven
Redundant altimeter	StratoLoggerCF
Primary deployment charge	3.5g Black powder
Redundant deployment charge	4g Black powder
Deployment altitude	1500 ft
Descent rate	6.58 m/s

Table 14: Shock cord and mechanical links information

Shock cord and mechanical links information	Values
Booster	
Shock cord length	8.22 m
Shock cord weight	72g
Shock cord material	Kevlar
Shock cord size	1/4 in
Number of quick links	4
Number of swivel links	2
Sustainer drogue	
Shock cord length	7.31 m
Shock cord weight	66g
Shock cord material	Kevlar
Shock cord size	1/4 in
Number of quick links	3
Number of swivel links	1
Sustainer main	
Shock cord length	14 m
Shock cord weight	124g
Shock cord material	Kevlar
Shock cord size	1/4 in
Number of quick links	5
Number of swivel links	1

Components	Qty	Weight (g)	Total Weight (g)
Avionics part 1	1	90	90
Avionics part 2	1	32	32
Avionics part 3	1	52	52
Avionics part 4	1	44	44
Arming PCB	1	110	110
RFD-900x	1	66	66
Blue Raven	1	6	6
Blue Jay+	1	6	6
ODB	1	52	52
Idefix	1	60	60
Cameras	1	32	32
Battery 18650	2	52	104
Main Battery Casing	1	180	180
Battery 21700	10	66	660
		Total	**1494**

Table 15: Component Mass Breakdown — Avionics Booster

Components	Qty	Weight (g)	Total Weight (g)
Structure part 1	1	90	90
Structure part 2	1	32	32
Structure part 3	1	52	52
Structure part 4	1	44	44
Arming PCB	1	110	110
RFD-900x	1	66	66
Blue Raven	1	6	6
Stratologger CF	1	20	20
ODB	1	52	52
Idefix	1	60	60
Cameras	1	32	32
Battery 18650	2	52	104
Main Battery Casing	1	180	180
Battery 21700	10	66	660
		Total	**1508**

Table 16: Component Mass Breakdown — Avionics Sustainer

Appendix B - Project Test Reports

CO2 Testing

This report summarizes the tests conducted on the CO2 deployment system designed for rocket separation and parachute deployment. The goal was to optimize system performance through iterative testing and adjustments.

Table 17: Summary of CO2 deployment system tests

Test #	Setup	Result
1	CO2 system alone	Successfull but e-match came out
2	New e-match configuration	Failed – broken O-ring and not enough black powder
3	Same as test 2	Success
4	New system design	Success
5	New system design	Success
6	Deployment test inside the rocket	Success rocket did separate

The CO2 system has had improvement since last year design. We made it more compact and had to change the dimension of certain parts during testing to improve the seal and the reliability.

Composite Testing

We did traction testing on our composite the get their strength and failure point in order to have more acurate calculations. Here are the final result of this testing.

Material	E_1 (GPa)	E_2 (GPa)	G_{12} (GPa)	ν_{12}	σ_{1t} (MPa)	σ_{2t} (MPa)	τ_{12f} (MPa)
E-glass 320 g plain weave	28.524	27.056	4.745	0.0845	600.33	420.31	64.58
CF 3k 2×2 twill 205 g	82.885	80.649	4.997	0.0283	934.95	962.24	66.76

Table 18: Experimentally determined ply properties used in laminate analysis.

Appendix C - Hazard Analysis

The following table enumerates the potential hazards that may be encountered and the measures taken to mitigate the risks.

Hazard	Possible Causes	Risk of Mishap	Mitigation Approach	Residual Risk
Motor explosion during launch	Grain cracks, structural failure, overpressure	Medium – limited testing and student manufacturing	Pressure testing of motor casing, visual inspection of grain, use of non-fragmenting materials, restricted safety perimeter during launch	Low
Stage separation failure	Improper charge sizing, mechanical misalignment, wiring failure	Medium – critical event with multiple failure points	Ground testing of separation system, redundant charges where possible, strict arming procedures	Low
Premature upper stage ignition	Avionics error, wiring fault, accidental arming	High – could occur near personnel	Safe/arm systems, continuity checks, ignition only enabled at final countdown	Low
Failure of upper stage ignition	Igniter failure, battery failure, timing error	Medium – mission failure but limited safety impact	Redundant ignition checks, battery testing, verified sequencing logic	Low
Recovery system failure	Improper packing, charge failure, altimeter malfunction	High – uncontrolled descent	Redundant deployment charges, ground testing, proper packing procedures	Medium
Rocket deviates from flight path	Aerodynamic instability, wind, CG/CP miscalculation	High – risk to personnel	Stability simulations, launch constraints (wind limits), safety exclusion zone	Low
Hang fire (delayed ignition)	Ignition delay or partial ignition	High – personnel may approach rocket	Strict launch protocol, minimum wait time before approach, remote ignition	Low
Pyrotechnic accidental ignition	Static discharge, wiring errors, premature connection	High – handling hazard	Shorting plugs, delayed connection, trained personnel only	Low
Exposure to composite dust	Sanding fiberglass/carbon fiber	Medium – health hazard over time	Use of respirators, designated ventilated workspace	Low
Black powder mishandling	Improper storage or transport	Medium – fire/explosion hazard	Storage in fire-resistant locked container, compliance with transport regulations	Low

Appendix D - Risk Assessment

The following table enumerates the potential risks that may be encountered and the measures taken to mitigate them.

Failure Mode	Mission Phase	Cause	Likelihood	Impact	Mitigation Approach
Ignition failure (stage 1)	Launch	Igniter failure, wiring issue	Low	High – no liftoff	Pre-launch continuity checks, igniter redundancy, battery verification
Thrust underperformance	Ascent	Improper grain casting, pressure loss	Medium	High – reduced altitude	Static motor testing, grain inspection, design margin
Aerodynamic instability	Ascent	CG/CP miscalculation, fin misalignment	Low	High – trajectory deviation	Stability simulation (OpenRocket), CG verification before launch
Stage separation failure	Stage separation	Charge failure, mechanical issue	Medium	High – upper stage unusable	Ground testing, redundant charges, robust mechanical design
Upper stage ignition failure	Stage separation	Igniter or avionics failure	Medium	High – mission incomplete	Redundant ignition checks, validated timing logic
Avionics malfunction	All phases	Software bug, wiring fault, power loss	Medium	High – loss of control/data	Code verification, secure wiring, battery testing
Altimeter failure	Recovery	Sensor failure, improper configuration	Medium	High – parachute not deployed	Redundant altimeters, pre-flight configuration check
Parachute deployment failure	Recovery	Packing error, charge failure	Medium	High – vehicle loss	Redundant deployment charges, packing verification
Excessive drift during descent	Recovery	High winds, late deployment	Medium	Medium – difficult recovery	Wind constraints, deployment altitude tuning
Structural failure during flight	Ascent	Material defect, excessive load	Low	High – mission loss	Structural analysis, safety factor in design

Appendix E - Checklists

Parachutes

Sustainer Drogue

Step	Done	Initials
Ensure proper folding of the sustainer drogue	☐	
Insert the cord into the nosecone	☐	
Wrap the sustainer drogue cord in an S-pattern using electrical tape	☐	
Attach and torque the sustainer drogue quicklink to the cord	☐	
Attach and torque the cord quicklink to the upper sustainer avionics bulkhead	☐	
Secure the cord inside the compartment	☐	
Attach and torque the cord quicklinks	☐	
Remove the tape around the drogue	☐	
Assemble the nosecone to the avionics coupler with 3 shear pins	☐	

Sustainer Main

Step	Done	Initials
Ensure proper folding of the sustainer main parachute	☐	
Wrap the sustainer main cord in an S-pattern using tape	☐	
Attach and torque the sustainer main quicklink to the cord	☐	
Attach and torque the quicklink to the sustainer motor bulkhead	☐	
Remove the tape around the main parachute	☐	
Insert the cord and main parachute into the tube	☐	
Attach and torque the cord quicklink to the lower sustainer avionics bulkhead	☐	
Assemble the tube to the avionics coupler with 3 shear pins	☐	

Booster Main and Drogue Harness

Step	Done	Initials
Ensure proper folding of the drogue parachute	☐	
Ensure proper folding of the main parachute	☐	
Insert the cord into the pressurization compartment	☐	
Wrap the sustainer drogue cord in an S-pattern using electrical tape	☐	
Attach and torque the sustainer drogue quicklink to the cord	☐	
Attach and torque the sustainer main quicklink to the cord	☐	
Attach and torque the cord quicklink to the the motor bulkhead	☐	
Attach and torque the cord quicklink to the the bottom booster avionic's bulkhead	☐	
Make sure the main parachute is properly secured to the shock cord with the tie wraps and the line cutters with proper slack in the heavy duty electrical wire	☐	
Insert the cord into the pressurization compartment	☐	
Remove the tape around the drogue	☐	
Insert the drogue and main into the pressurization compartment	☐	
Assemble the booster avionic's tube to the motor tube with 3 shear pins	☐	

Avionics Bay

Sustainer Avionics Bay

Step	Done	Initials
Screw the black powder charges onto the bulkhead	☐	
Screw the CO2 systems onto the bulkhead	☐	
Activate the spot tracker	☐	
Connect the primary main chute e-match to the Blue Raven main channel	☐	
Connect the primary drogue e-match to the Blue Raven apogee channel	☐	
Connect the redundant main chute e-match to the Stratologger main channel	☐	
Connect the redundant drogue e-match to the Stratologger apogee channel	☐	
Connect the motor igniter to the arming PCB terminal block	☐	
Verify all device connections on the arming PCB	☐	
Verify all battery connections on the arming PCB	☐	
Verify connections on all altimeters	☐	
Verify mounting of all components and secure with tape or zip ties if needed	☐	
Connect the igniter wire to the lower sustainer avionics bulkhead using electrical tape	☐	
Insert and slide the sustainer bay into the avionics tube	☐	
Tighten the 4 M5 screws of the upper sustainer avionics bulkhead	☐	
Connect the igniter wire with the Wago connector between the bulkhead and the pressurization compartment	☐	
Mount the camera module on rails with the external cap	☐	
Test igniter continuity	☐	
Test e-match continuity	☐	

Booster Avionics Bay

Step	Done	Initials
Screw the black powder charges onto the bulkhead	☐	
Screw the CO2 systems onto the bulkhead	☐	
Connect the primary main chute e-match to the Blue Raven main channel	☐	
Connect the primary drogue e-match to the Blue Raven apogee channel	☐	
Connect the redundant main chute e-match to the Blue Jay Plus main channel	☐	
Connect the redundant drogue e-match to the Blue Jay Plus apogee channel	☐	
Connect the separation charge e-match to Blue Raven channel 3	☐	
Verify all device connections on the arming PCB	☐	
Verify all battery connections on the arming PCB	☐	
Verify connections on all altimeters	☐	
Verify mounting of all components and secure with tape or zip ties if needed	☐	
Insert and slide the booster bay into the avionics tube	☐	
Tighten the 4 M5 screws of the upper booster avionics bulkhead	☐	
Mount the camera module on rails with the external cap	☐	
Test e-match continuity	☐	

Motors

Sustainer

Step	Done	Initials
Verify installation of the eyebolt on the sustainer motor bulkhead	☐	
Route the sustainer igniter through a conduit	☐	
Screw the sustainer motor bulkhead into the fuselage with 4 M5 screws	☐	
Install an e-match (fake igniter on the wire)	☐	
Tape the protruding wire section to the outside of the motor tube	☐	
Insert the channel plug piece around the motor tube	☐	
Tape the excess string to the outside of the motor tube	☐	
Screw the sustainer motor into the eyebolt	☐	

Booster

Step	Done	Initials
Tighten the 6 axial M5 screws of the booster motor bulkhead	☐	
Tighten the retainer screws	☐	
Insert the booster motor	☐	
Screw the retainer	☐	
Tape the booster igniter to the outside of the tube	☐	
Verify tightness of the rail button screws	☐	

Pad

All-up Test

Step	Done	Initials
Install the sustainer upright on the all-up stand	☐	
Keep the test igniter away from the rocket	☐	
Wear safety glasses and gloves	☐	
Insert push-pin 2 to power the Blue Raven	☐	
Insert push-pin 3 for the shunt	☐	
Test continuity	☐	
Remove the push-pins	☐	
Lay the sustainer horizontally on supports on the table	☐	
Verify igniter attachment at two points on the stick	☐	
Insert the stick into the printed cap	☐	
Install the cap on the nozzle using electrical tape	☐	
Secure excess igniter wire with tape on the side of the motor	☐	

Pad

Step	Done	Initials
Assemble both stages	☐	
Slide the rail buttons into the launch rail	☐	
Raise and secure the launch rail	☐	
Wear safety glasses	☐	
Climb the ladder to booster height	☐	
Insert the three booster push-pins	☐	
Climb the ladder to sustainer height	☐	
Insert the three sustainer push-pins	☐	
Confirm RFD900 GPS lock with the ground station	☐	
Confirm Featherweight GPS tracker lock	☐	
Test radio communication between Featherweight GPS tracker and base station	☐	
Insert the booster motor igniter	☐	
Install ground camera	☐	
Arm the Featherweight devices	☐	

Appendix F - Engineering Drawings

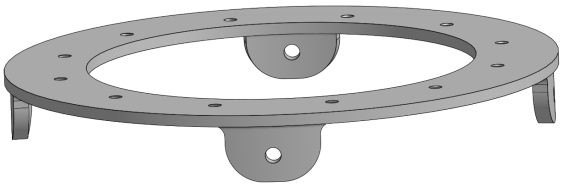


Fig. 23 CAD of the booster's thrustplate

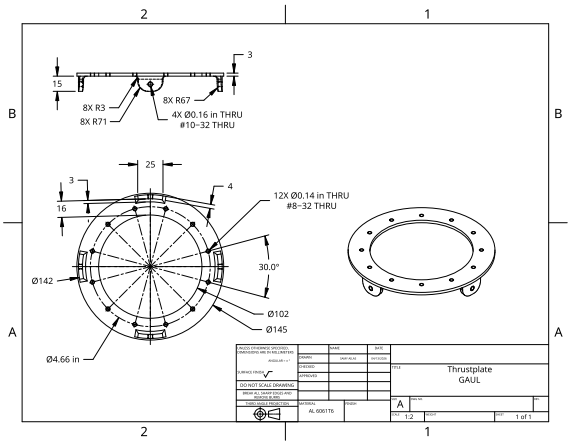


Fig. 24 Engineering drawing of the booster's thrustplate

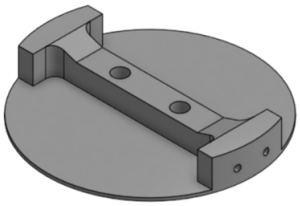


Fig. 25 CAD of the booster's motor bulkhead

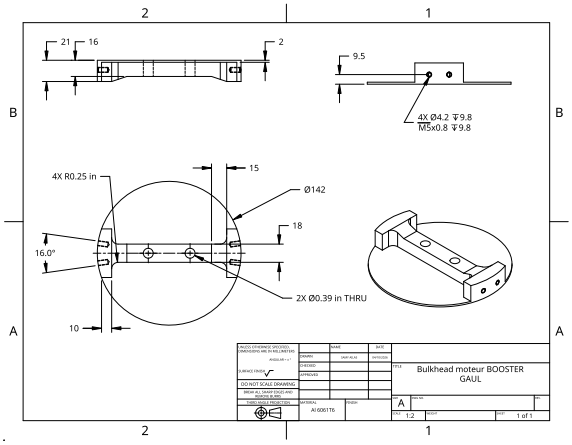


Fig. 26 Engineering drawing of the Booster motor bulkhead

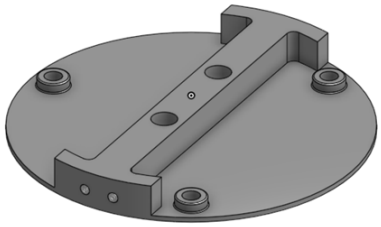


Fig. 27 CAD of the booster bottom avionic bay bulkhead

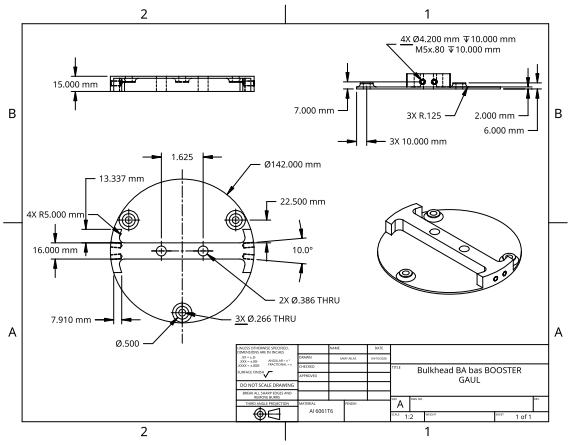


Fig. 28 Engineering drawing of booster's bottom avionic bay bulkhead

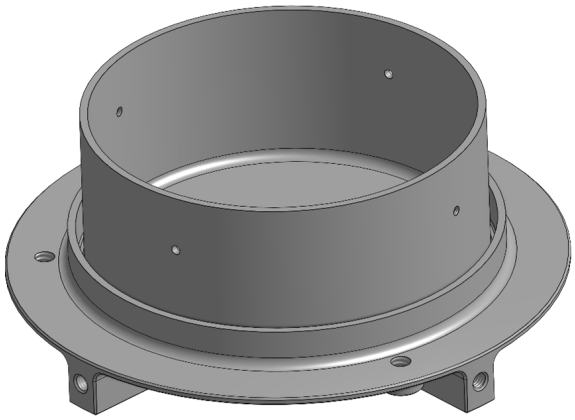


Fig. 29 CAD of the booster's top avionic bay bulkhead

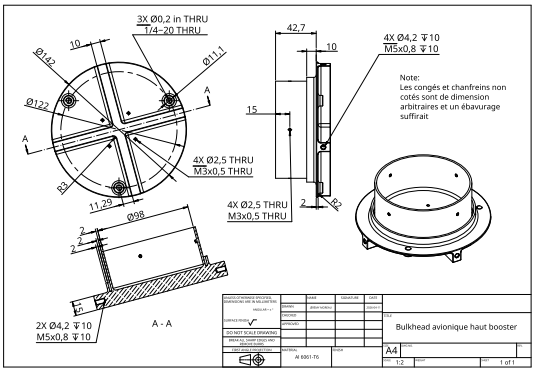


Fig. 30 Engineering drawing of the booster's top avionic bay bulkhead

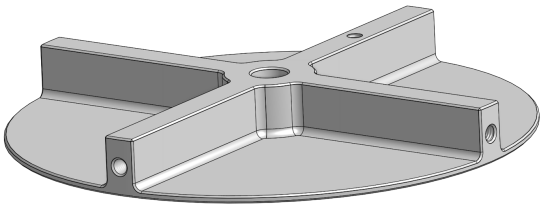


Fig. 31 CAD of the sustainer's motor bulkhead

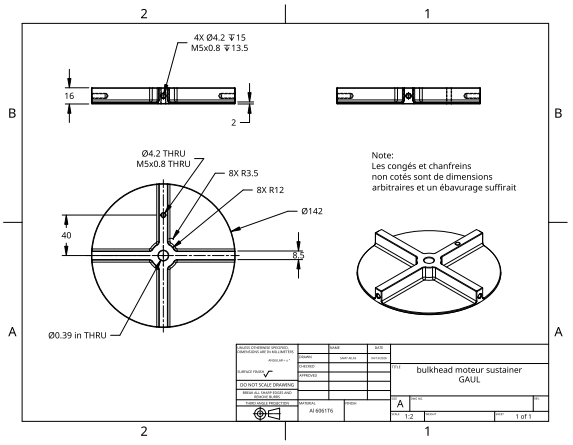


Fig. 32 Engineering drawing of the sustainer's motor bulkhead

Payload CubeSat Drawings

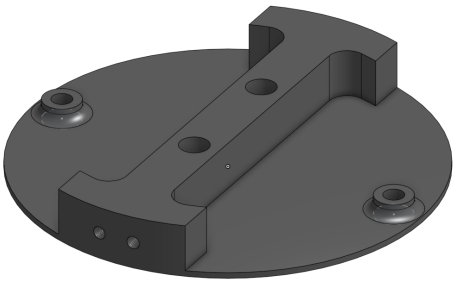


Fig. 33 CAD of the sustainer's bottom avionic bay bulkhead

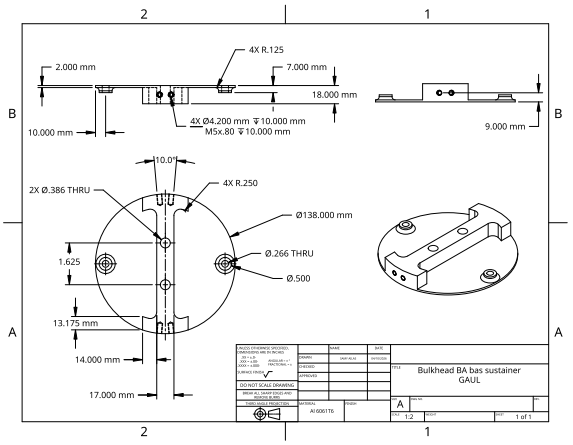


Fig. 34 Engineering drawing of the sustainer's bottom avionic bay bulkhead

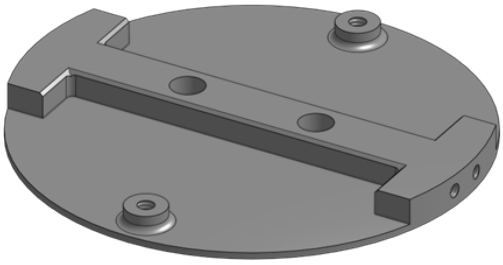


Fig. 35 CAD of the sustainer's top avionic bay bulkhead

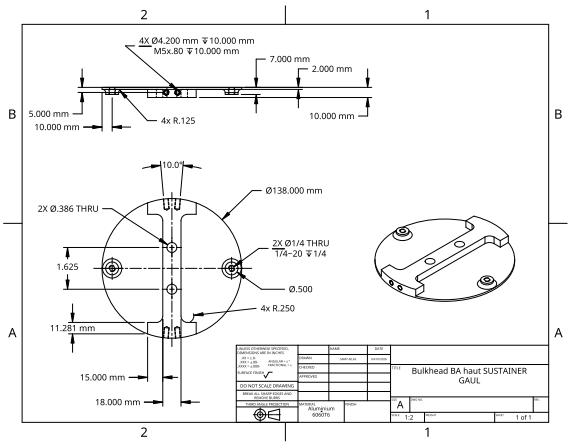


Fig. 36 Engineering drawing of the sustainer's top avionic bay bulkhead

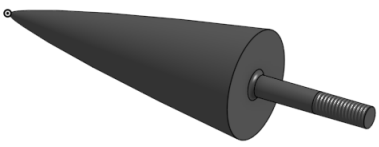


Fig. 37 CAD of the nose tip

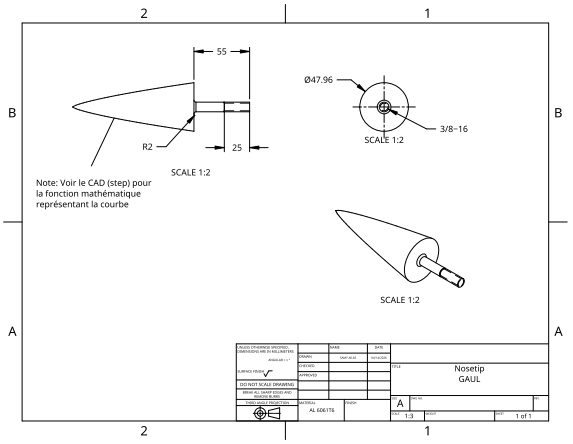


Fig. 38 Engineering drawing of the nose tip

Appendix G - Flight simulations

The flight performance prediction methodology, simulation tools, and flight dynamics analysis approach were introduced in the System Architecture section and are further detailed in this appendix. Flight simulations

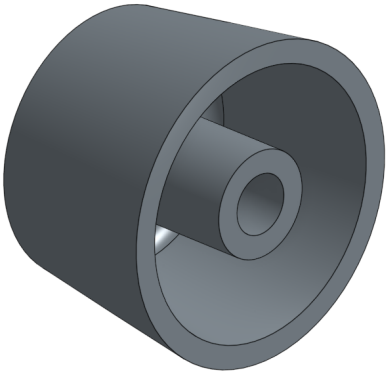


Fig. 39 CAD of the nose tip's base

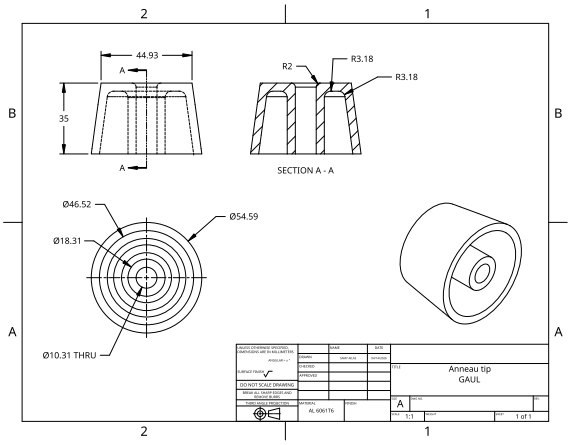


Fig. 40 Engineering drawing of the nose tip's base

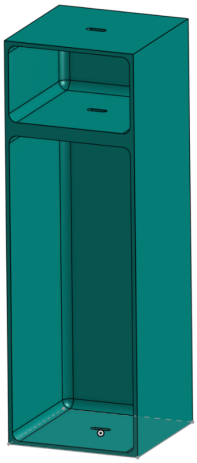


Fig. 41 CubeSat drawing

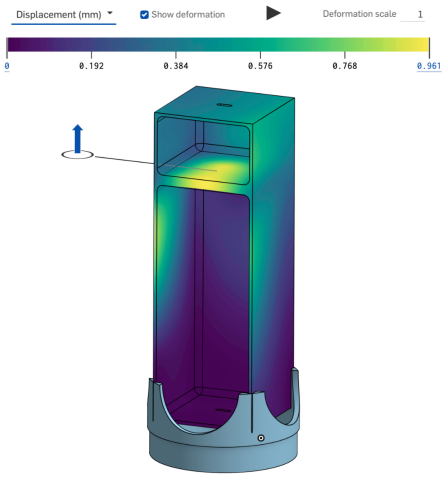


Fig. 42 CubeSat Deformation Simulation

were conducted using four independent methods: OpenRocket, RasAero, RocketPy, and the internally developed SRAD software SimulOutil, presented in the following appendix.

The use of multiple tools is intentional and serves two purposes. First, it provides cross-validation of key outputs such as apogee altitude, velocity history, stability margin, and landing dispersion. Second, discrepancies between models help identify the dominant uncertainties affecting the vehicle, particularly for a complex two-stage rocket operating through transonic and supersonic regimes.

OpenRocket is the primary tool used for general trajectory analysis, including altitude, velocity, acceleration, and overall stability trends throughout the flight. RasAero is primarily used when more refined aerodynamic and center of pressure predictions are required, making it the reference tool for detailed static stability assessments. RocketPy provides an additional physics-based trajectory model with expanded three-dimensional flight dynamics capabilities. SimulOutil is used as an internally controlled verification method and as a platform for sensitivity studies.

Ascent Kinematics

The first performance metric considered is the ascent kinematics of the two primary powered-flight configurations: the full stack during booster propulsion and the sustainer during second-stage ascent. These plots provide the predicted evolution of altitude, velocity, and acceleration throughout the mission.

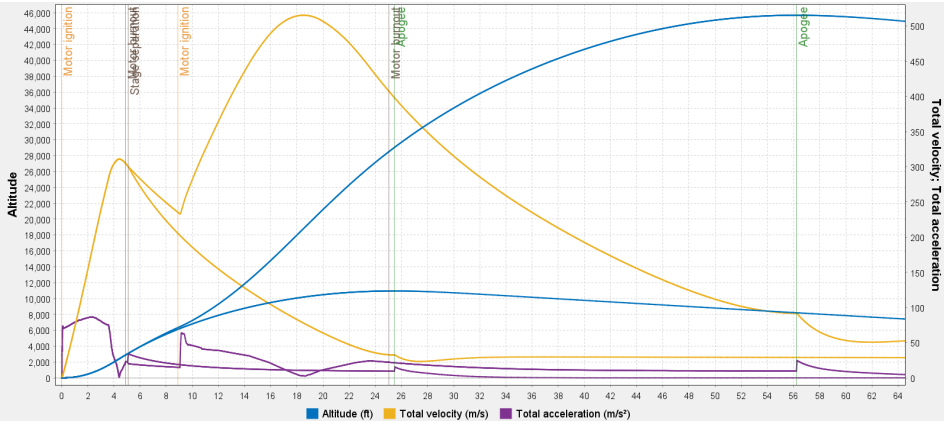


Fig. 43 Predicted kinematics for the complete ascent phase.

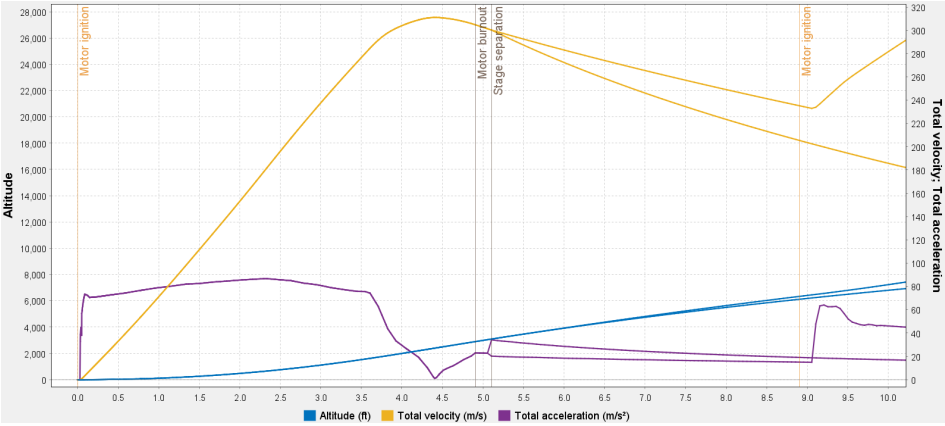


Fig. 44 Predicted kinematics during booster-powered stacked ascent.

These results are used to verify maximum acceleration, maximum velocity, burnout conditions, coast duration, staging timing, and expected apogee. A three-dimensional trajectory can also be reconstructed from RocketPy outputs, allowing visualization of lateral drift and vertical progression throughout ascent.

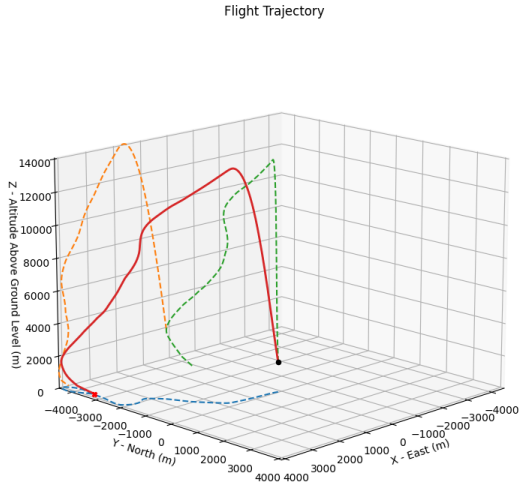


Fig. 45 Three-dimensional simulated trajectory from RocketPy.

Static Stability Margins

Ascent stability is evaluated using two complementary metrics: caliber margin, which is the team's preferred internal metric, and stability margin expressed as a percentage of total rocket length, which corresponds to the competition requirement.

The required minimum stability margin is 7.5% throughout the subsonic regime and 10% throughout the supersonic regime. In addition to these requirements, internal targets were established to maintain at least 3 calibers for the full stack and 2 calibers for the sustainer throughout ascent.

The percentage-based metric ensures rule compliance, while caliber margin remains a practical measure for comparing relative static stability between flight configurations.

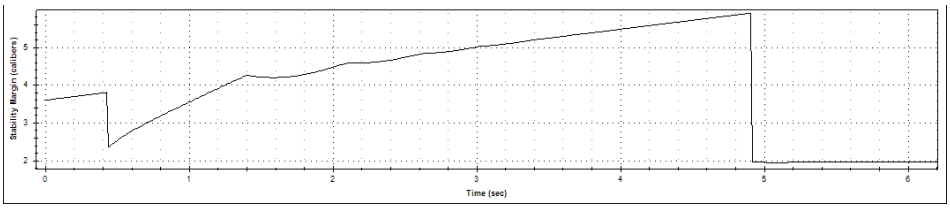


Fig. 46 Full-stack stability margin predicted with RasAero.

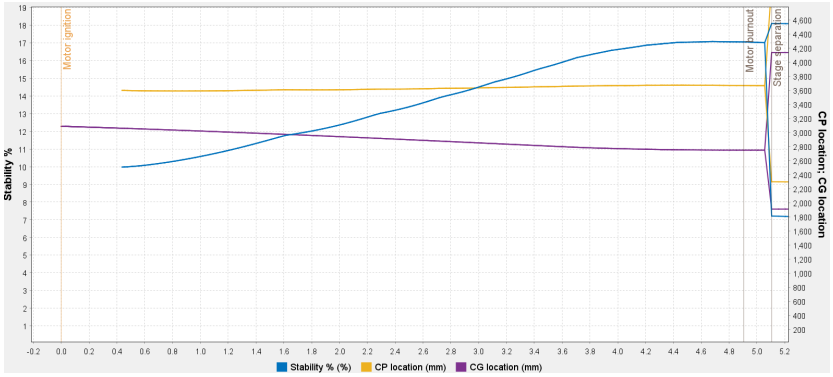


Fig. 47 Full-stack stability margin, center of gravity, and center of pressure predicted with OpenRocket.

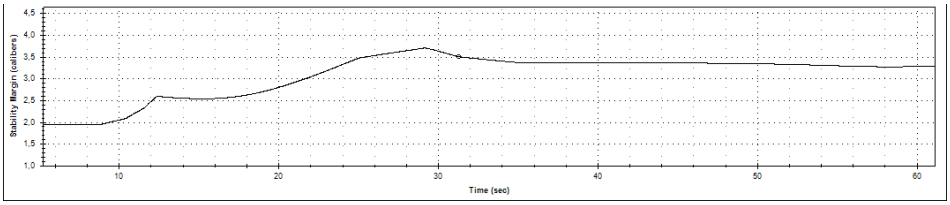


Fig. 48 Sustainer stability margin predicted with RasAero.

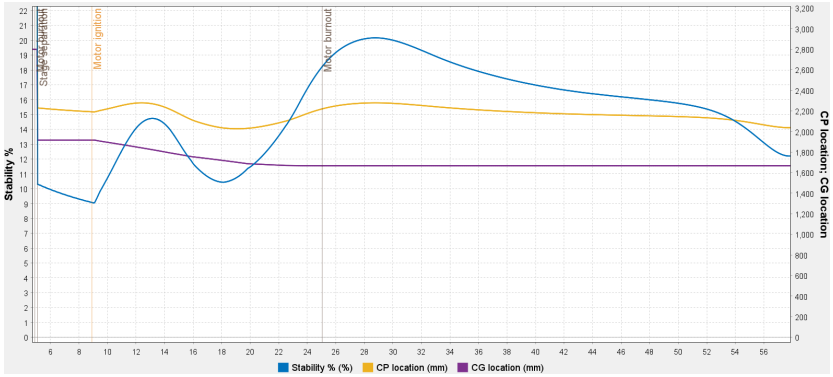


Fig. 49 Sustainer stability margin, center of gravity, and center of pressure predicted with OpenRocket.

Thrust-to-Weight Ratio

Another critical parameter verified through simulation is the thrust-to-weight ratio history, with particular emphasis on lift-off conditions. This metric governs initial acceleration, launch rail exit velocity, and aerodynamic control authority immediately after departure.

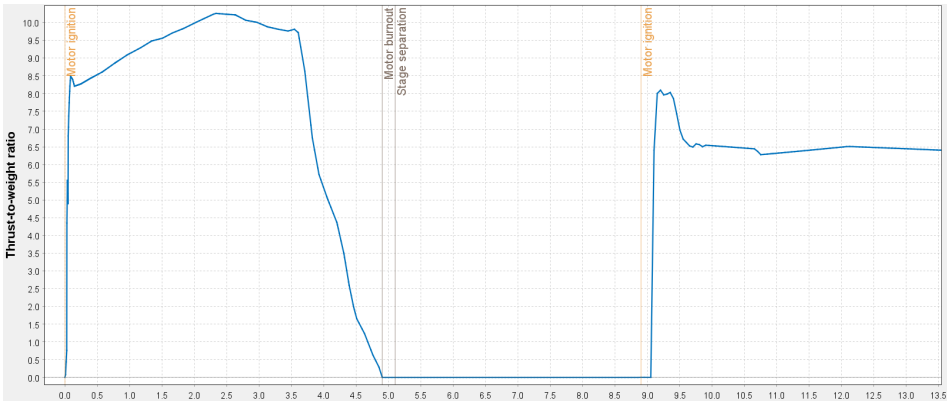


Fig. 50 Predicted thrust-to-weight ratio evolution throughout powered flight.

The design requirement is to exceed a ratio of 8 at launch for safe rail departure and robust initial ascent and it satisfies this requirement for the four first seconds.

Upper-Stage Ignition Attitude

A parameter specific to multistage rockets is in-flight ignition of the sustainer motor, which occurs without rail guidance. The vehicle must therefore remain as close as possible to the vertical axis at ignition in order to comply with ignition inhibition safety requirements.

The angle from vertical at sustainer ignition is consequently a critical quantity. Excessive deviation may reduce trajectory efficiency, increase lateral range, and challenge safe ignition criteria.

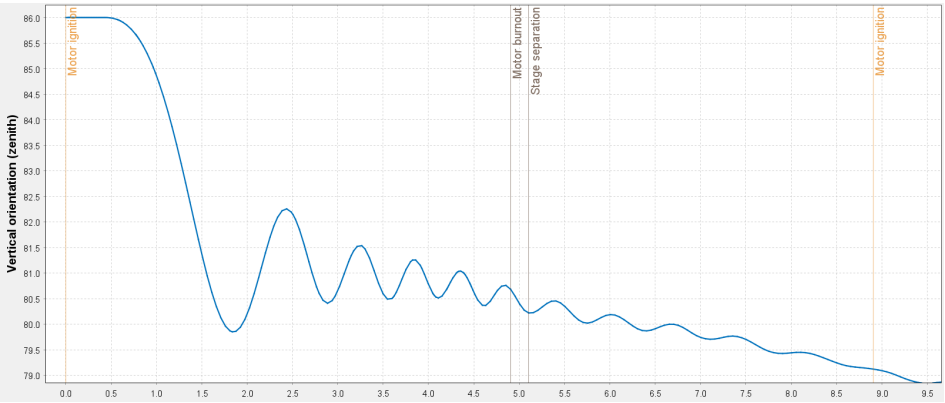


Fig. 51 Predicted rocket orientation relative to the horizon during flight.

Particular attention is therefore given to coast-phase attitude evolution between stage separation and sustainer ignition.

Launch Conditions

Launch site atmospheric conditions have a significant effect on trajectory behavior and recovery drift. The environmental assumptions used in the simulations are summarized below. Wind data were modeled using a multi-altitude profile obtained from OpenMeteo, providing a more representative input than a single ground-level wind value.

Characteristic	Value
Site altitude (ft)	2950
Temperature (°C)	35
Air pressure (kPa)	91
Wind profile	Multi-level (Fig. 52)
Effective rail length (ft)	18.5
Rail angle from vertical (deg)	4

Table 28: Launch conditions used for simulations.

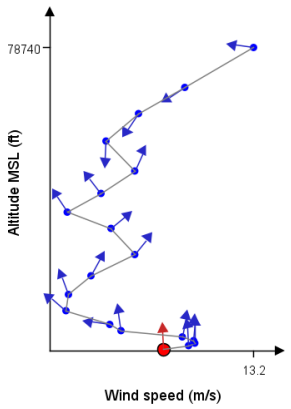


Fig. 52 Wind profile used in trajectory and landing simulations.

Apogee Prediction Comparison

The final predicted altitude remains one of the most critical mission parameters. Results from the four independent methods are compared below.

Source	OpenRocket	RasAero	RocketPy	SimulOutil
Apogee altitude (ft)	45,600	53,400	46,100	48,400

Table 29: Comparison of predicted apogee altitude between simulation tools.

The spread in predictions confirms the importance of using multiple methods for a complex two-stage mission.

Model-to-Model Diagnostic Comparisons

To better understand differences between simulation outputs, four fundamental time histories were compared: acceleration, thrust, drag, and mass evolution. These parameters directly govern the trajectory solution and help isolate disagreement sources.

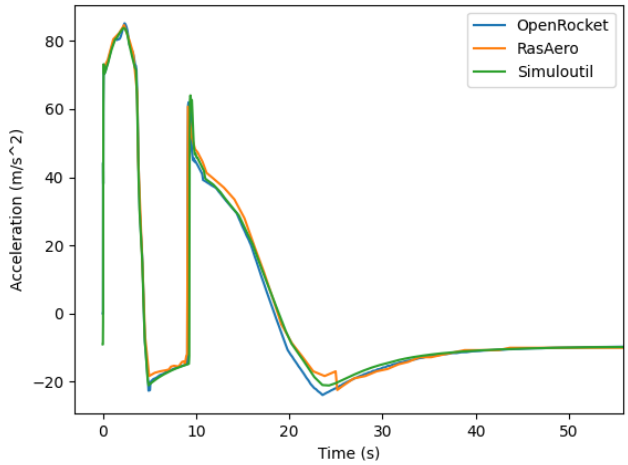


Fig. 53 Comparison of acceleration histories from multiple methods.

During booster flight, agreement is strong. Larger discrepancies appear during sustainer burn and especially at its burnout where velocity is highest and aerodynamic sensitivity is greatest.

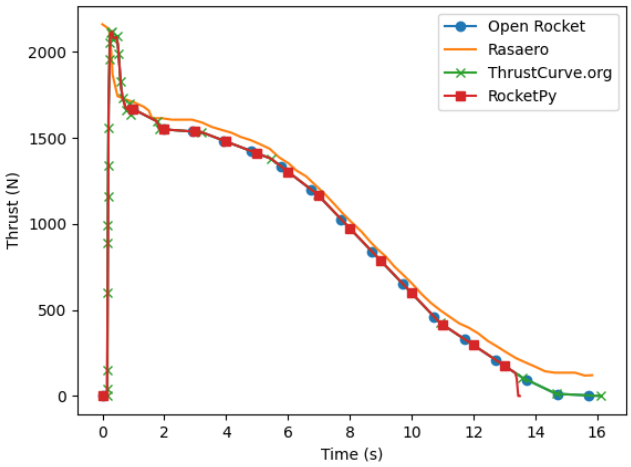


Fig. 54 Comparison of N1000 thrust curves used by different methods.

Small differences in the sustainer thrust curve may contribute to divergence in final altitude predictions, particularly when compounded over the entire burn duration. The effect of more thrust on the sustainer motor could be an important source of the prediction diferences between RasAero and the others.

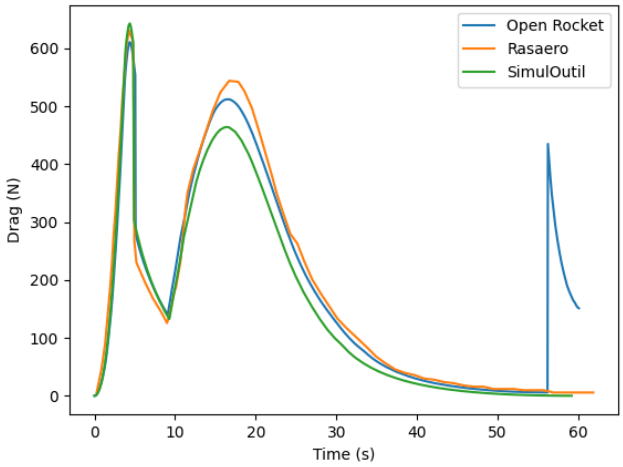


Fig. 55 Comparison of drag force histories during flight.

The largest differences occur near maximum velocity, confirming that supersonic drag modeling is the most critical parameters requiring refinement. Other than in the supersonic regime, the three calculated drag forces agree.

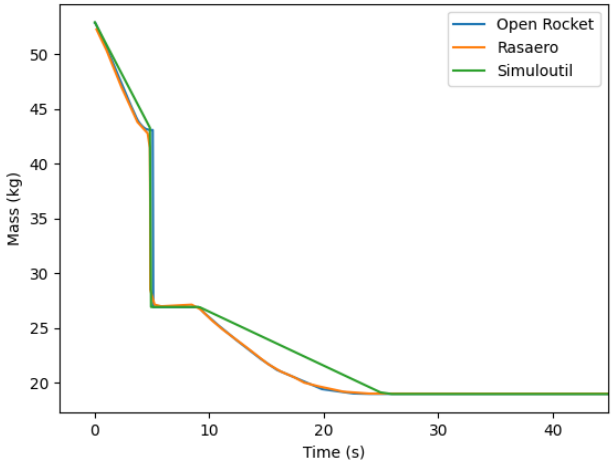


Fig. 56 Comparison of propellant mass evolution assumptions.

The mass comparison indicates that the current linear propellant consumption model used in SimulOutil is a gross simplification. Future improvements may include non-linear burn mass evolution derived directly from thrust curve behavior to improve acceleration fidelity.

Appendix H - SimulOutil

Simuloutil is a centralized analysis environment developed to address the difficulty of predicting complex rocket flight behavior using only commercial off-the-shelf (COTS) softwares. While COTS tools provide valuable baseline simulations, they often lack flexibility for custom modeling, detailed subsystem analysis, and mission-specific scenarios. To overcome this limitation, additional in-house analysis tools have been developed and integrated into a single Python-based framework. The goal of Simuloutil is to combine these modules into one coherent environment that can be used alongside COTS softwares. The seven currently developed modules are explained in detail below.

1 - Flight and Trajectory

The trajectory tool is an internally developed flight simulation tool whose purpose is to provide an apogee prediction to compare result with other methods like OpenRocket, RasAero and RocketPy.

Calculations:

The simulation is based on Newton's second law applied to the rocket at each time step:

$$\frac{T - D - W}{m} = a \tag{5}$$

Velocity and altitude are then obtained by integration of the acceleration:

$$v = \int a \, dt, \qquad h = \int v \, dt \tag{6}$$

The aerodynamics drag force is calculated as:

$$D = \frac{1}{2} \rho \, A_{\text{ref}} \, C_D(v) \, v^2 \tag{7}$$

The drag coefficient C_D is directly interpolated from a C_D vs mach number curve. The velocity and air density calculated with altitude are computed at each time step and used to find the next values in the iteration, thus needing adequate time step selection. The Motor thrust is also interpolated from the thrust-curve passed as input. Mass is estimated as linearly decreasing during the burn. For the two-stage simulations, the booster and sustainer are solved sequentially, with the final condition of the booster used as the initial condition for the sustainer.

The two-stage calculations is extended to three dimensions. The vector associated with a time step includes the position, velocity and acceleration all in the up, north and east directions, allowing to simulate drift due to wind throughout the flight.

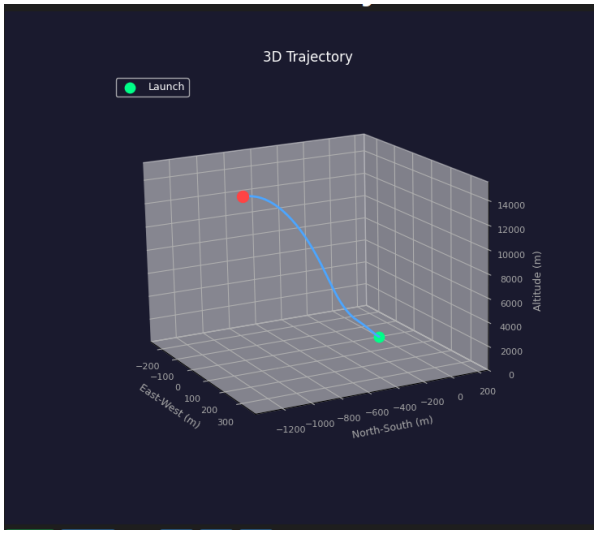


Fig. 57 An example of the 3D trajectory calculated with SimulOutil

2 - Aerodynamic (C_D Calculation)

The aerodynamic module computes the drag coefficient as a function of Mach number. These results can then be compared with values obtained from computational fluid dynamics (CFD) simulations, OpenRocket, RasAero and Aerolab to validate the accuracy and reliability of the methodology.

The aerodynamic model considers parasite drag as the dominant contribution, assuming negligible lift effects due to the rocket's near-axisymmetric geometry and small angle of attack. Parasite drag is decomposed into the following components:

1. Friction drag, caused by viscous shear stresses along the rocket surface

2. Pressure drag, resulting from flow separation and shape effects

3. Base drag, generated by the low-pressure wake

4. Wave drag, which becomes significant in transonic and supersonic regimes due to shock formation

5. Interference drag, arising from interactions between components

Each of these contributions is estimated using empirical equations derived from experimental data and established aerodynamic correlations. These equations relate the drag coefficient to parameters such as Mach number, Reynolds number, and geometric properties of the vehicle. The total drag coefficient is obtained by summing all individual contributions, providing a practical and computationally efficient approximation of aerodynamic behavior across different flight regimes.

The tool supports both single-stage and two-stage rocket configurations, allowing users to analyze and compare different flight architectures. Detailed plots are provided for each individual component of drag (friction, pressure, base, wave, and interference), enabling a deeper understanding of their respective contributions across the flight profile. Furthermore, a continuously updated table displays key parameters throughout the simulation, including a Mach number indicator bar, providing a clear and intuitive representation of the flight regime.

The following graphic illustrates a typical graph that can be obtained with the Aerodynamic module.

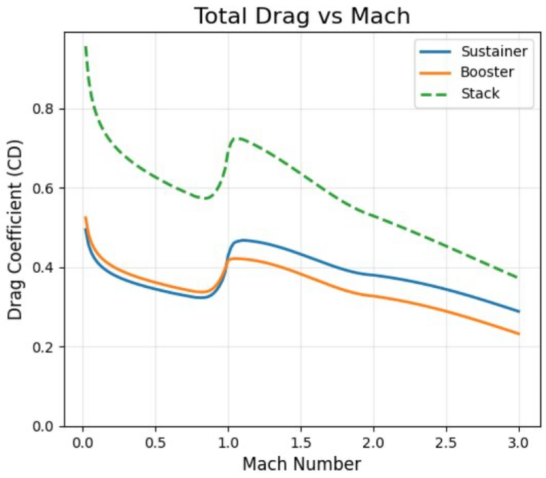


Fig. 58 Drag coefficient in function of the Mach number

3 - Aerodynamic analysis (comparison)

The aerodynamic analysis module provides a flexible framework for evaluating drag using multiple modeling approaches. It allows users to import external drag coefficient (C_D) data for validation and comparison, while offering customizable plotting tools to visualize selected curves. This enables efficient analysis and a deeper understanding of aerodynamic performance across different conditions. The following image illustrates the module's interface.

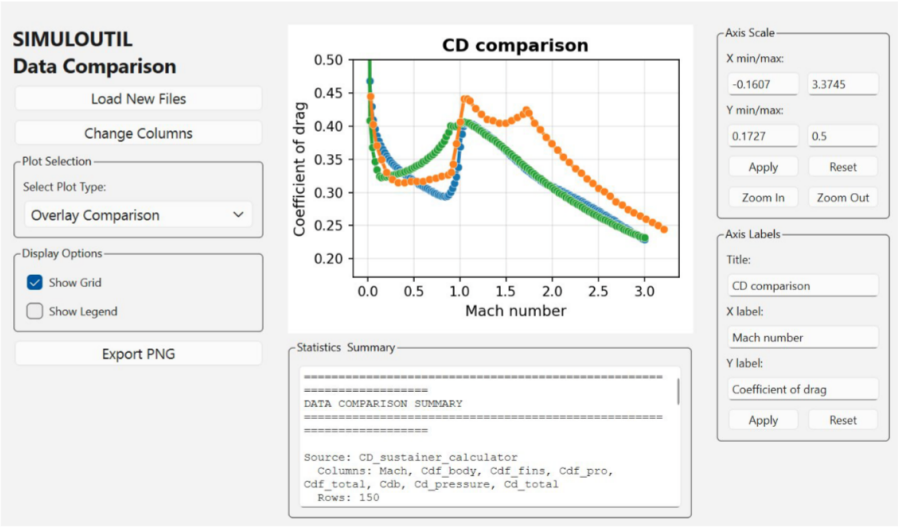


Fig. 59 Aerodynamic Analysis User Interface

4 - Landing Dispersion

The landing dispersion module predicts the rocket's landing position using simplified physics-based modeling of the descent phase after apogee. It accounts for parachute and vehicle surface areas, aerodynamic drag on both the rocket and deployed parachute, and the relative airflow velocity vector. Starting from apogee conditions, the equations of motion are integrated over time to simulate the descent under gravity and drag forces. By repeating this simulation multiple times with historical wind data and varying atmospheric conditions, the model generates a statistical dispersion pattern, providing a realistic estimate of potential landing zones and touchdown variability.

The following graphic illustrates a resulting landing scatter on a map.

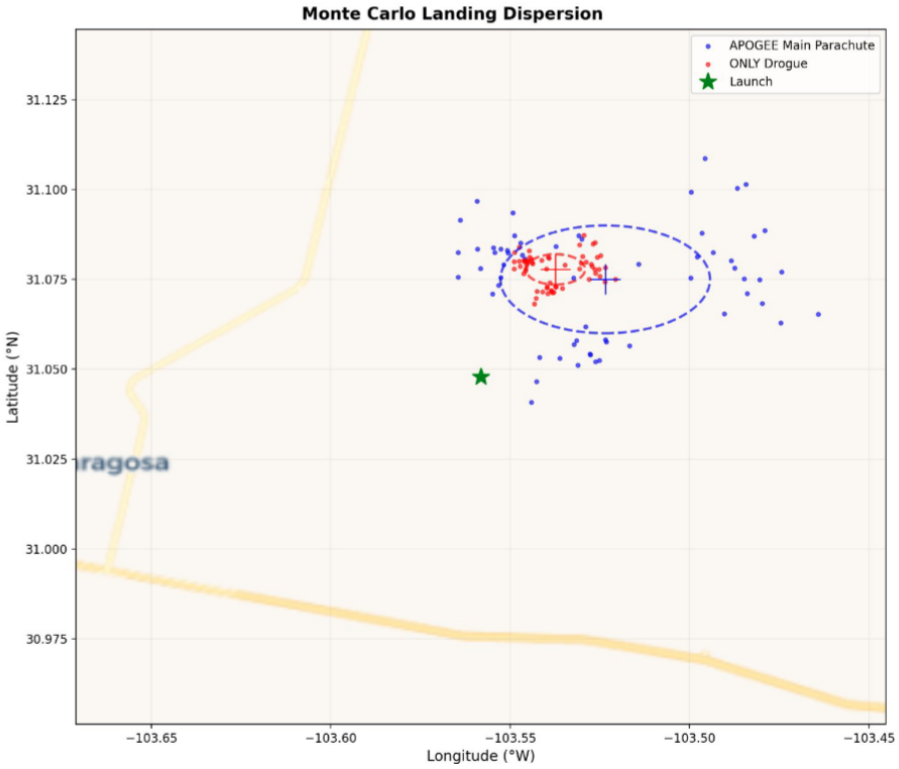


Fig. 60 Landing Scatter on a Map

5 - Flight Structural Load

The load analysis module evaluates the structural loads and stresses acting on the airframe throughout flight by considering the most critical aerodynamic and inertial conditions. Two main cases are studied. We suppose that a wind gust orients the rocket at a 10 degrees angle of attack while at maximum dynamic pressure (Max Q), generating bending loads driven by lift forces, aerodynamic side loads, and inertial reactions along the structure. In this scenario, bending results from the combined effect of aerodynamic lift and the vehicle's resistance to motion changes.

In the second case, axial loading during powered ascent is analyzed, where the airframe experiences compression due to thrust, opposed by aerodynamic drag and modified by inertial reactions from acceleration.
Together, these two cases allow the module to identify peak bending moments and maximum axial stresses, ensuring the structural integrity of the rocket under both aerodynamic and propulsion-driven loading conditions. Examples of graphics produced by this module are illustrated in the following appendix I.

6 - Composite materials design

The composite module uses classical laminate theory to calculate safety factors from different criterion (such as Tsai-Hill, Tsai-Wu, max stress, and max strain), deformations, and maximum stress in different sandwich and ply layups. It also calculates such characteristics in individual plies throughout the composite and indicates the most likely point(s) of failure. An example of the equations and uses of the module are demonstrated in the following appendix I with the calculations made directly for our composites.

7 - Fin Flutter

The module determines the critical velocity at which aeroelastic flutter is likely to occur. This is achieved using the following equation:

$$V_f = a \times \sqrt{\frac{G_E}{\left(\frac{39.3 \times A^3}{\left(\frac{t}{c}\right)^3 \times (A+2)}\right) \times \left(\frac{\lambda+1}{2}\right) \times \left(\frac{p}{p_0}\right)}} \tag{8}$$

It features an intuitive and user-friendly interface where the geometrical parameters, material properties and flight conditions are entered. When updated, a geometrical representation of the fin appears to the side and helps to mitigate mistakes in the geometrical parameters often confusing.

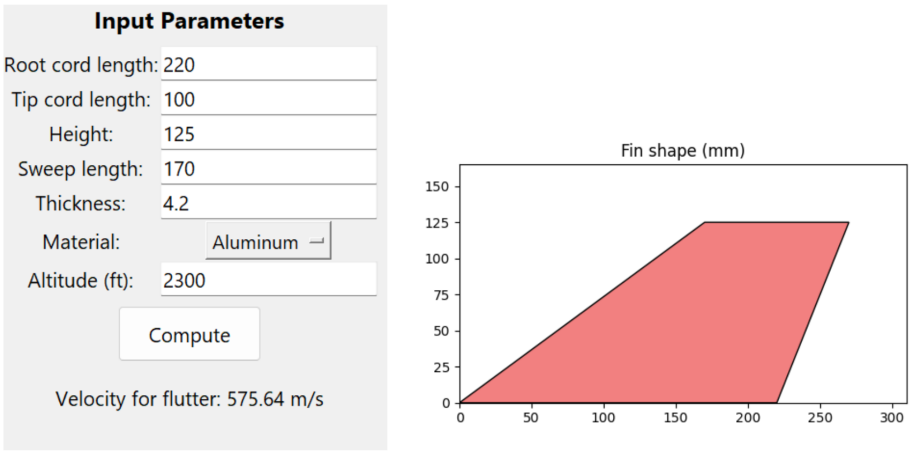


Fig. 61 Fin Flutter Module User Interface

Appendix I - Airframe calculations

Material Characterization Mechanical properties for both the E-glass and carbon fibre laminates were determined experimentally via uniaxial tensile testing of SRAD coupons conducted in the laboratory. The following orthotropic ply properties were extracted and used as inputs to all subsequent analyses:

Material	E_1 (GPa)	E_2 (GPa)	G_{12} (GPa)	ν_{12}	σ_{1t} (MPa)	σ_{2t} (MPa)	τ_{12f} (MPa)
E-glass 320 g plain weave	28.524	27.056	4.745	0.0845	600.33	420.31	64.58
CF 3k 2×2 twill 205 g	82.885	80.649	4.997	0.0283	934.95	962.24	66.76

Table 30: Experimentally determined ply properties used in laminate analysis.

Load Cases Structural loads are evaluated at the most critical flight instant: immediately before motor burnout, where thrust, velocity, and aerodynamic stability margin are simultaneously near their maximum. Two configurations are analysed independently: the full stack (booster + sustainer) and the sustainer alone after stage separation.

At this instant, an aerodynamic disturbance induces a 10° angle of attack, causing the rocket to resist the off-axis flow with both asymmetric drag and inertial restoring forces. This produces a distributed bending moment along the airframe superimposed on the axial compressive load from motor thrust. The resulting load envelope for each configuration is shown in Figure 62.

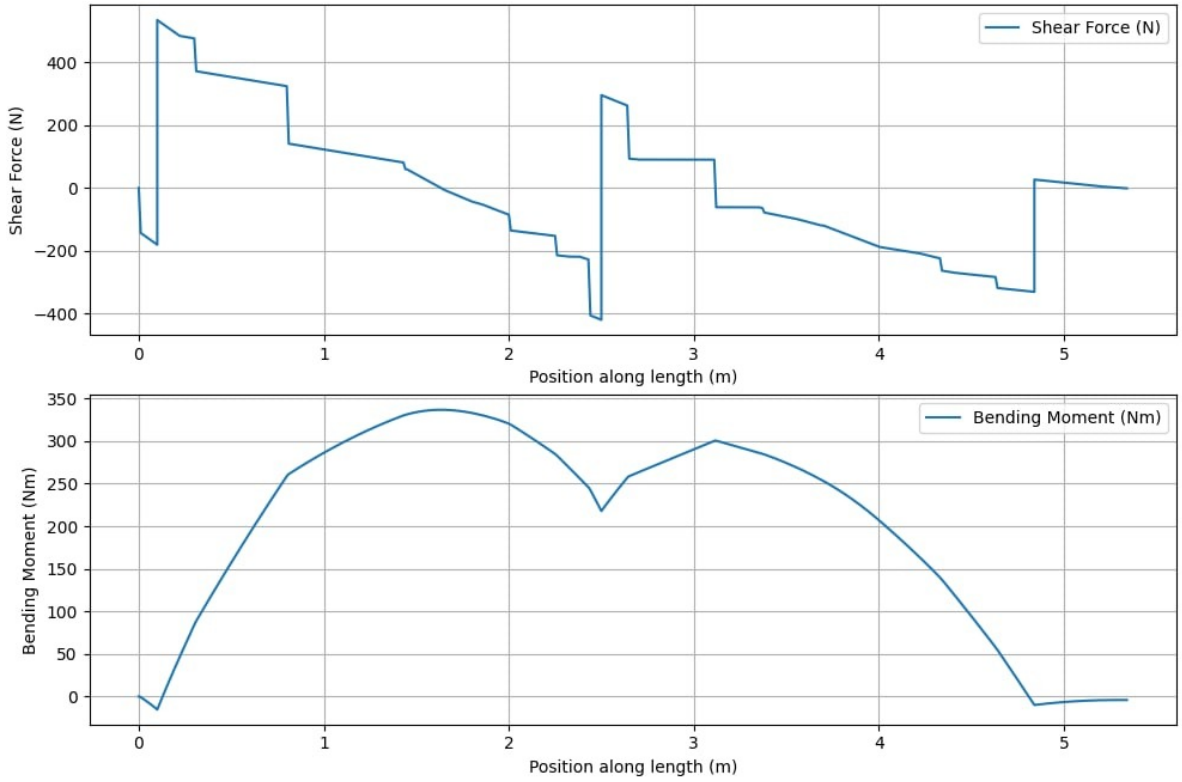


Fig. 62 Distributed bending moment along the airframe for the full-stack and sustainer-only load cases at the critical flight instant.

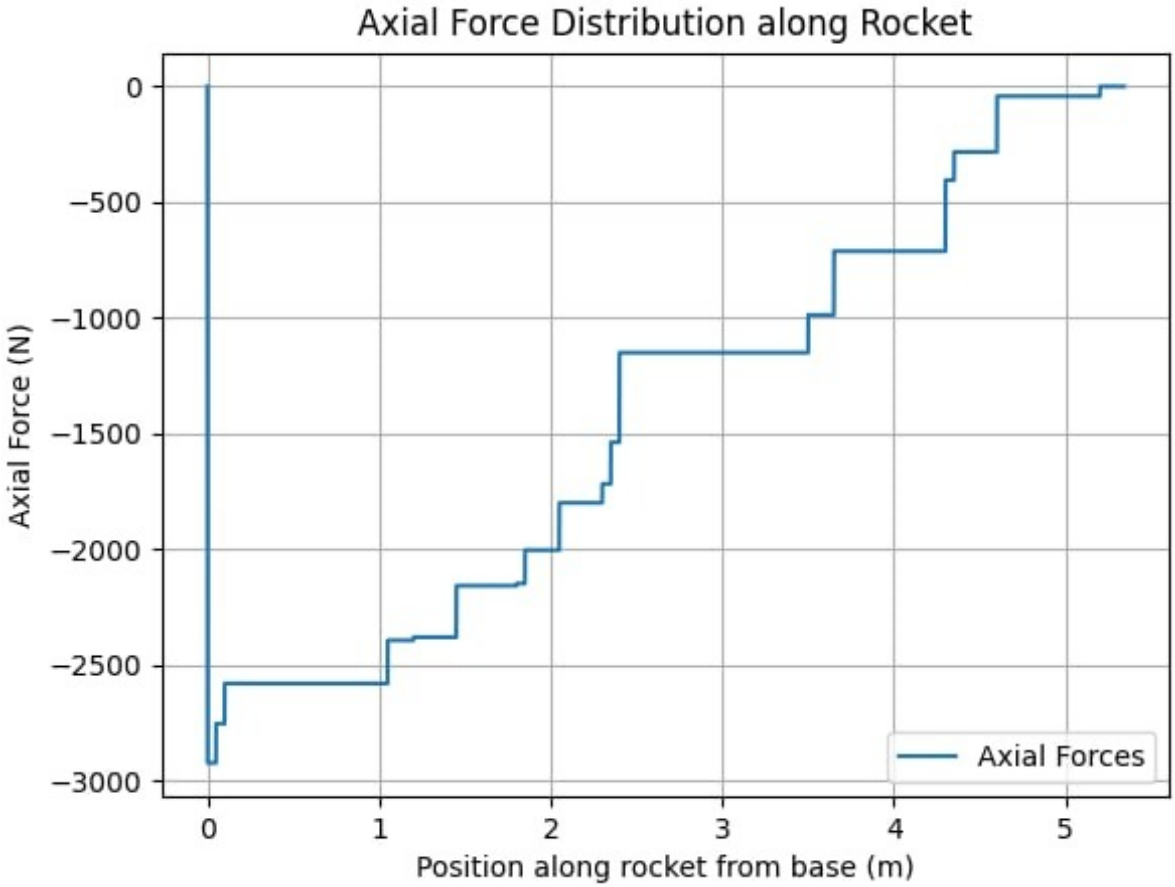


Fig. 63 Distributed axial force along the airframe for the full-stack and sustainer-only load cases at the critical flight instant.

Laminate Stress Analysis Stresses are resolved at the ply level using Classical Laminate Theory (CLT) with the Kirchhoff plate hypothesis, which assumes that normals to the mid-plane remain straight and normal after deformation. This yields the through-thickness strain field:

$$\begin{Bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{Bmatrix} = \begin{Bmatrix} \varepsilon_x^0 \\ \varepsilon_y^0 \\ \gamma_{xy}^0 \end{Bmatrix} + z \begin{Bmatrix} \kappa_x \\ \kappa_y \\ \kappa_{xy} \end{Bmatrix} \tag{9}$$

where ε^0 are the mid-plane strains and κ are the curvatures, both obtained by inverting the ABD constitutive relation:

$$\begin{Bmatrix} \mathbf{N} \\ \mathbf{M} \end{Bmatrix} = \begin{bmatrix} A_{11} & A_{12} & A_{16} & B_{11} & B_{12} & B_{16} \\ A_{12} & A_{22} & A_{26} & B_{12} & B_{22} & B_{26} \\ A_{16} & A_{26} & A_{66} & B_{16} & B_{26} & B_{66} \\ B_{11} & B_{12} & B_{16} & D_{11} & D_{12} & D_{16} \\ B_{12} & B_{22} & B_{26} & D_{12} & D_{22} & D_{26} \\ B_{16} & B_{26} & B_{66} & D_{16} & D_{26} & D_{66} \end{bmatrix} \begin{Bmatrix} \varepsilon_x^0 \\ \varepsilon_y^0 \\ \gamma_{xy}^0 \\ \kappa_x \\ \kappa_y \\ \kappa_{xy} \end{Bmatrix} \tag{10}$$

The ABD matrix is assembled from the individual ply reduced stiffness matrices $\bar{Q}_{ij}$, themselves derived from the experimentally measured ply properties (E_1, E_2, G_{12}, ν_{12}) and the ply orientation angles. The full construction procedure is implemented in Python.

Applied Loads The worst-case loading on a tube section combines axial compression from motor thrust and compression from the bending moment, both acting on the same generator line. The corresponding in-plane running loads are:

$$N_x^{\text{axial}} = \frac{F_{\text{axial}}}{\pi d} \tag{11}$$

$$N_x^{\text{bending}} = \frac{32\, M\, d\, t}{\pi\left[(d+t)^4 - d^4\right]} \tag{12}$$

$$N_x^{\text{total}} = N_x^{\text{axial}} + N_x^{\text{bending}} \tag{13}$$

where d is the inner diameter, t the total laminate thickness, F_{axial} the net axial force, and M the local bending moment at the section of interest.

Ply-Level Stress and Failure Analysis Ply stresses in the fibre coordinate system (σ_1, σ_2, τ_{12}) are recovered by applying the transformation matrix to the laminate strains at each ply mid-plane. Four failure criteria are evaluated simultaneously: Maximum Stress, Maximum Strain, Tsai–Hill, and Tsai–Wu. The stress distribution through the laminate thickness and the resulting failure indices for each ply and criterion are shown in Figures 64 and 65.

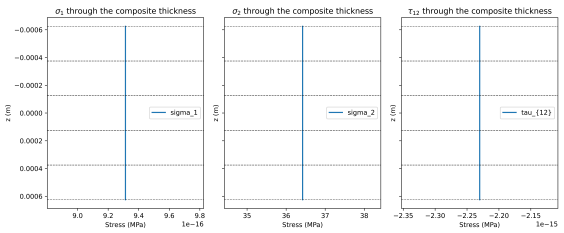


Fig. 64 Through-thickness ply stress distribution (σ_1, σ_2, τ_{12}) for the critical tube under worst-case loading.

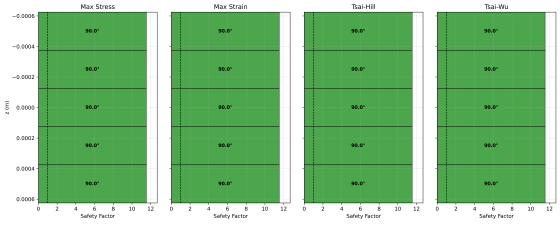


Fig. 65 Ply-level safety factors for all four failure criteria (Maximum Stress, Maximum Strain, Tsai–Hill, Tsai–Wu).

Buckling Analysis Local buckling of the tube wall under axial compression is evaluated for the two structurally critical cases: the worst-case carbon fibre tube and the worst-case E-glass tube. The critical buckling load is computed and compared against the applied compressive running load N_x^{total}.

Appendix J - Internal structure calculations

This subsection of the appendix details the structural calculations for the internal structure components, using both analytical material mechanics and finite element analysis (FEA) sections (III.A.2 and III.C.1.

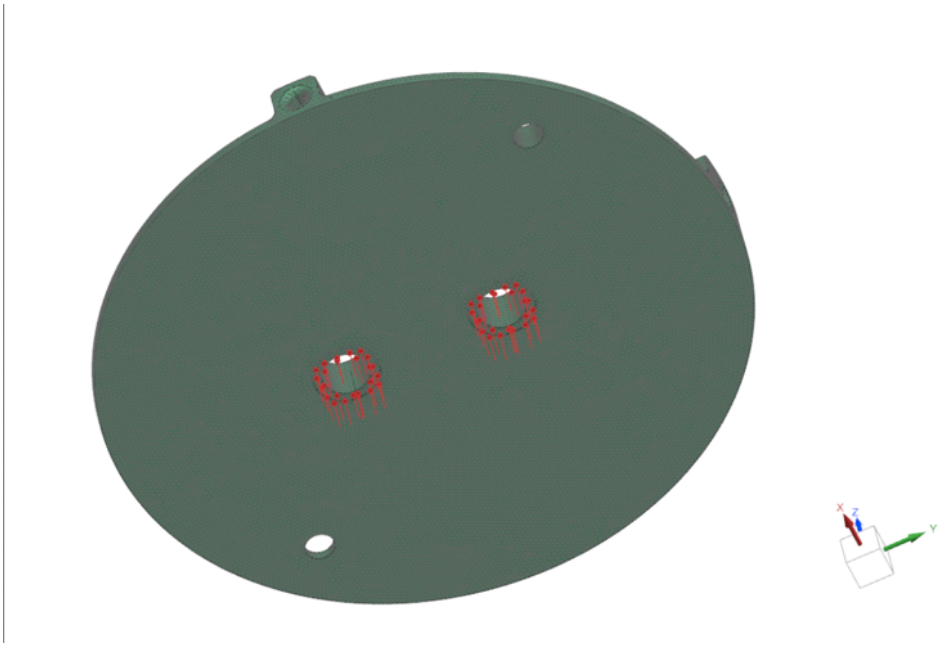


Fig. 66 Applied force on the bulkheads with U-bolts on FEA

Thrusplate

The thrust plate is loaded in two distinct phases: the boost phase, where the motor thrust is applied to the plate, and the parachute deployment, where the empty motor weight pulls on the opposite side.

Anticipated loads

During the boost phase, the load is the product of the maximum rocket acceleration and the mass of the stack above the plate:

$$F_{\text{boost}} = ma = 52 \text{ kg} \cdot 8.04\ G \cdot 9.81 \text{ m/s}^2 = 4100 \text{ N} \tag{14}$$

For the parachute deployment, a conservative case is assumed with a 20 G deceleration and the empty motor mass:

$$F_{\text{deploy}} = ma = 6.57 \text{ kg} \cdot 20\ G \cdot 9.81 \text{ m/s}^2 = 1290 \text{ N} \tag{15}$$

Shear stress (boost phase)

The shear stress is obtained by dividing the load by the area in the shear plane:

$$\tau = \frac{F}{A} = \frac{F}{\pi \cdot d \cdot e} = \frac{4100 \text{ N}}{\pi \cdot 142 \text{ mm} \cdot 3 \text{ mm}} = 3 \text{ MPa} \tag{16}$$

Bending stress (boost phase)

The bending moment is obtained from Roark's formulas for circular flat plates, using the case of external edges fixed and central portion free:

$$M = w \cdot a \left(L_9 - \frac{C_7 L_6}{C_4} \right) \tag{17}$$

where the coefficients are defined as:

$$L_9 = \frac{r_0}{a}\left(\frac{1+\nu}{2}\ln\left(\frac{a}{r_0}\right) + \frac{1-\nu}{4}\left(1-\left(\frac{r_0}{a}\right)^2\right)\right) \tag{18}$$

$$L_6 = \frac{r_0}{4a}\left(\left(\frac{r_0}{a}\right)^2 - 1 + 2\ln\left(\frac{a}{r_0}\right)\right) \tag{19}$$

$$C_4 = \frac{1}{2}\left((1+\nu)\frac{b}{a} + (1-\nu)\frac{a}{b}\right) \tag{20}$$

$$C_7 = \frac{1}{2}(1-\nu^2)\left(\frac{a}{b} - \frac{b}{a}\right) \tag{21}$$

with b the inner radius, a the outer radius, r_0 the radial position of the load, ν Poisson's ratio and w the applied load.

Using $b = 51$ mm, $a = 72.5$ mm, $r_0 = 51$ mm, $\nu = 0.33$ and $w = 4100$ N, the coefficients are $L_9 = 0.224$, $L_6 = 0.035$, $C_4 = 0.944$ and $C_7 = 0.32$, yielding $M = 63{,}000$ Nmm.

The bending stress follows from

$$\sigma = \frac{My}{I} = \frac{My}{\frac{bh^3}{12}} = \frac{63 \times 10^3 \text{ Nmm} \cdot 1.5 \text{ mm}}{\frac{440 \text{ mm} \cdot (3 \text{ mm})^3}{12}} = 95.45 \text{ MPa} \tag{22}$$

where y is the maximum distance to the neutral axis and I is the second moment of area of a rectangular section of width equal to the circumference and height equal to the plate thickness.

Safety factor

Bending is the critical loading. The resulting safety factor is

$$SF = \frac{S_y}{\sigma} = \frac{276 \text{ MPa}}{95.45 \text{ MPa}} = 2.89 \tag{23}$$

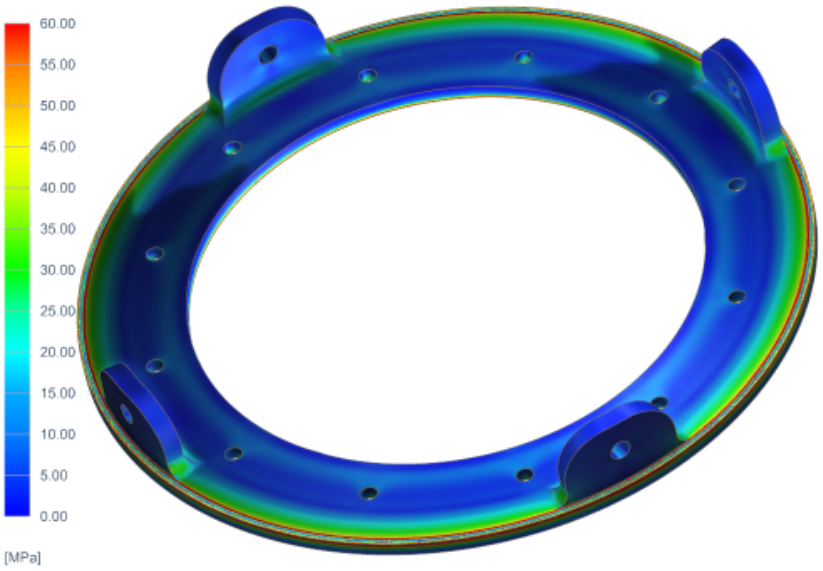


Fig. 67 Isometric view of the thrusplate finite element analysis

SUSTAINER - motor bulkhead

This bulkhead serves three functions: transmitting the sustainer motor thrust, transmitting the booster thrust to the sustainer, and anchoring the main parachute. Three load cases are evaluated, each corresponding to one function.

Anticipated loads

The sustainer motor thrust is read directly from the thrust curve at $T = 2100$ N.

The booster-to-sustainer thrust transmission is computed from the maximum boost acceleration applied to the sustainer mass:

$$F_{\text{boost}} = ma = 29 \text{ kg} \cdot 8\ G \cdot 9.81 \text{ m/s}^2 = 2275 \text{ N} \tag{24}$$

For the parachute deployment, a conservative case assumes the main parachute fully inflates instantly while the rocket is travelling at drogue descent speed and the motor failed to ignite. The total drag force is

$$D = \tfrac{1}{2}\rho v^2 C_D A = \tfrac{1}{2} \cdot 1 \text{ kg/m}^3 \cdot (30.7 \text{ m/s})^2 \cdot 1.55 \cdot 3.58 \text{ m}^2 = 2615 \text{ N} \tag{25}$$

The fraction acting on the bulkhead is determined by mass distribution on each side of the shock cord:

$$F_{\text{deploy}} = \frac{m_{\text{motor}}}{m_{\text{total}}} \cdot D = \frac{17.15 \text{ kg}}{29.1 \text{ kg}} \cdot 2615 \text{ N} = 1540 \text{ N} \tag{26}$$

The dimensioning load is therefore the booster thrust transmission at $F = 2275$ N.

Bending stress

The bulkhead is modeled as a beam fixed at both ends with a point load applied at its center, considering only the central transversal bars. The shear force and bending moment diagram yields a maximum bending moment of $M = 80.8 \times 10^3$ Nmm.

The second moment of area for the rectangular cross-section is

$$I = \frac{bh^3}{12} = \frac{18 \text{ mm} \cdot (16 \text{ mm})^3}{12} = 6.144 \times 10^3 \text{ mm}^4 \tag{27}$$

The bending stress is then

$$\sigma = \frac{My}{I} = \frac{80.8 \times 10^3 \text{ Nmm} \cdot 8 \text{ mm}}{6.144 \times 10^3 \text{ mm}^4} = 105.2 \text{ MPa} \tag{28}$$

FEA validation

A finite element analysis was performed in Siemens NX to confirm the analytical results. The maximum von Mises stress is approximately 80 MPa near the center, slightly below the hand calculation as expected from the conservative assumptions, and in the same order of magnitude. This corresponds to a safety factor of

$$SF = \frac{S_y}{\sigma_{\text{FEA}}} = \frac{276 \text{ MPa}}{80 \text{ MPa}} = 3.45 \tag{29}$$

against the 6061-T6 yield strength. Stress concentrations observed at the bolt holes are non-physical artifacts caused by the fixed-edge boundary conditions and are not representative of the actual stress state.

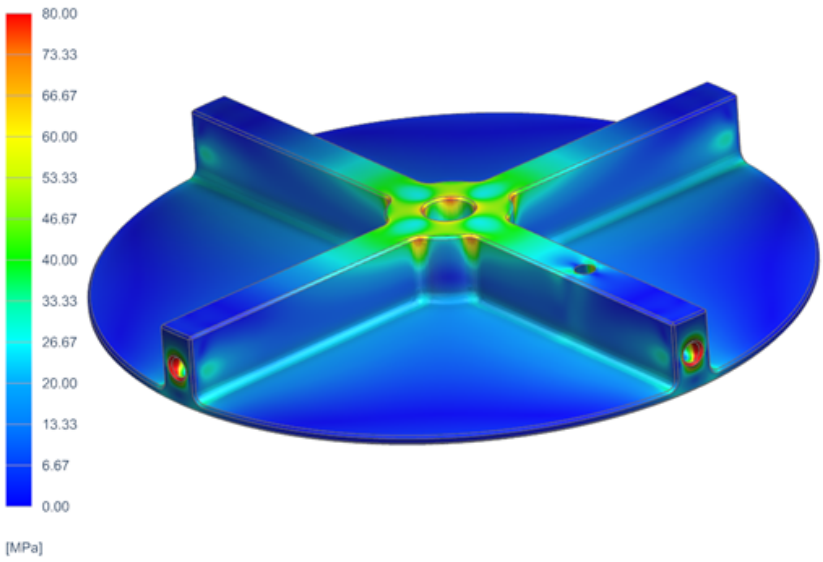


Fig. 68 Isometric view of the sustainer's motor bulkhead finite element analysis

BOOSTER - motor bulkhead

The booster motor bulkhead is subjected to two distinct load cases during flight. The first is the motor thrust load during the boost phase, transmitted axially through the motor tube bond and reacted by the bulkhead into the airframe. The second, and dimensioning, load case is the parachute snatch force at apogee deployment, which is transmitted through the shock cord into the U-bolt attachments and then into the bulkhead structure. A finite element analysis was performed using Siemens NX Simulation to determine the stress and strain distribution across the bulkhead under the maximum anticipated recovery load of 3116N. The bulkhead geometry was meshed and boundary conditions were applied to the M5 screw holes, with the recovery load applied at the U-bolt hole locations. Three critical regions were identified: The upper half of the U-bolt holes experiences a maximum stress of 52 MPa, yielding a safety factor of 5.29. The lower U-bolt holes are loaded with a maximum stress of 59 MPa and a corresponding safety factor of 4.66. Both values are well above the minimum safety factor of 3.0 targeted for recovery-critical structural components. The axial M5 fasteners represent the most critically loaded elements of the assembly, with a maximum stress of 147 MPa and a safety factor of 1.87. While this value falls below the 3.0 target applied to the bulkhead plate itself, it remains acceptable given the conservative assumptions used in the simulation.

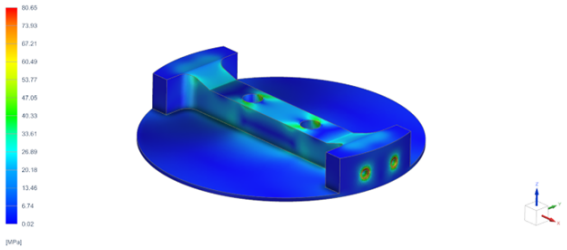


Fig. 69 Isometric view of the booster's motor bulkhead finite element analysis

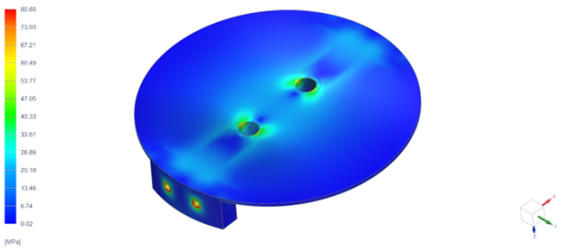


Fig. 70 Under isometric view of the booster's motor bulkhead finite element analysis

BOOSTER - Avionic bay bottom bulkhead

The lower avionics bay bulkhead is primarily loaded during parachute deployment, when the recovery snatch force is transmitted through the shock cord into the U-bolt attachments and then into the bulkhead structure. A finite element analysis was performed in Siemens NX Simulation to determine the stress distribution under the maximum anticipated recovery load. The bulkhead geometry was meshed with boundary conditions applied to the M5 screw holes and the recovery load applied at the U-bolt hole locations. Three critical regions were identified. The upper U-bolt holes experience a maximum stress of 40.25 MPa, yielding a safety factor of 6.83, while the lower U-bolt holes are loaded to 31.25 MPa with a corresponding safety factor of 8.8. Both values are well above the minimum safety factor of 3.0 targeted for recovery-critical structural components. The axial M5 fasteners represent the most critically loaded elements of the assembly, with a maximum stress of 117.11 MPa and a safety factor of 2.34. While this value falls below the 3.0 target applied to the bulkhead plate itself, it remains acceptable given the conservative assumptions used in the simulation.

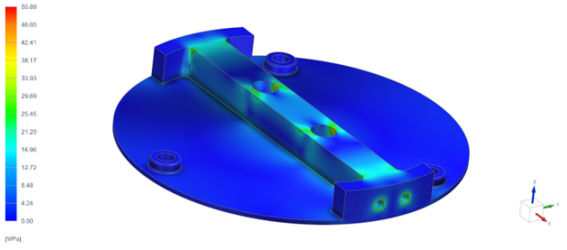


Fig. 71 Isometric view of the booster's bottom avionic bay bulkhead finite element analysis

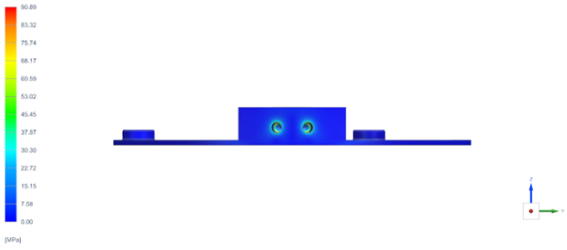


Fig. 72 Front view of the booster's bottom avionic bay bulkhead finite element analysis

BOOSTER - Avionic bay top bulkhead

This bulkhead is the structural core of the interstage, transferring the booster thrust to the sustainer through the sustainer motor that rests on its central cylinder. It is fixed to the fuselage with four radial M5 bolts and supports the interstage tube on its upper cylindrical section. A single dimensioning load case is considered: the booster thrust transmission during the boost phase.

Anticipated load

The load corresponds to the sustainer mass accelerated at the maximum stack acceleration of 8 G:

$$F = ma = 27\ \text{kg} \cdot 8\ G \cdot 9.81\ \text{m/s}^2 = 2120\ \text{N} \tag{30}$$

Bending stress

The bulkhead is modeled as a beam, and a shear force diagram yields the maximum bending moment:

$$M = 1060\ \text{N} \cdot \left(21 + \frac{50}{2}\right)\ \text{mm} = 48.76 \times 10^3\ \text{Nmm} \tag{31}$$

The approximated second moment of area of the beam is

$$I = \frac{bh^3}{12} = \frac{20\ \text{mm} \cdot (15\ \text{mm})^3}{12} = 5.63 \times 10^3\ \text{mm}^4 \tag{32}$$

The bending stress is then

$$\sigma = \frac{My}{I} = \frac{48.76 \times 10^3 \text{ Nmm} \cdot 7.5 \text{ mm}}{5.63 \times 10^3 \text{ mm}^4} = 64.9 \text{ MPa} \tag{33}$$

FEA validation

A finite element analysis was performed in Siemens NX to confirm the analytical results and obtain more precise stress values. The maximum stress is approximately 40 MPa, located at two spots: the lower border of the central cylinder and the radial bolt holes. Compared to the aluminum yield strength of 275 MPa, this yields a safety factor of

$$SF = \frac{S_y}{\sigma_{\text{FEA}}} = \frac{275 \text{ MPa}}{40 \text{ MPa}} = 6.88 \tag{34}$$

This is well above the 3.0 target, so the design could theoretically be further optimized for mass. However, manufacturing and ease-of-assembly constraints justify this margin.

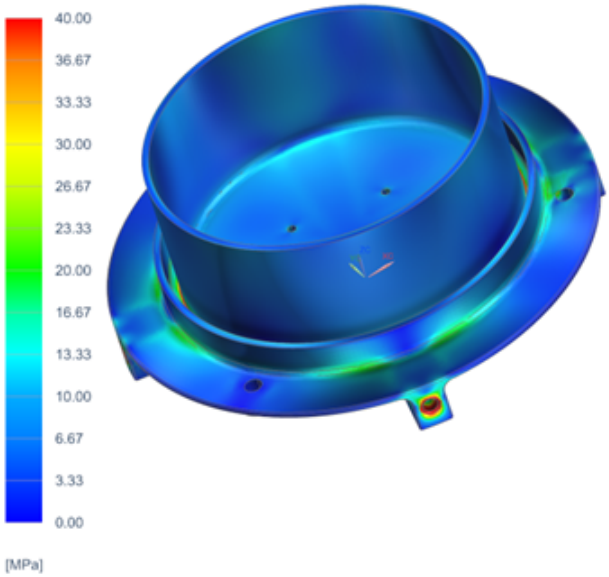


Fig. 73 Isometric view of the booster's top avionic bay bulkhead finite element analysis

SUSTAINER - Avionic bay bottom bulkhead

A finite element analysis was performed in Siemens NX Simulation with boundary conditions applied to the M5 screw holes and the recovery load applied at the U-bolt hole locations. Three critical regions were identified. The upper and lower U-bolt holes experience maximum stresses of 60.17 MPa and 56.99 MPa respectively, both well above the minimum safety factor of 3.0 targeted for recovery-critical structural components. The axial M5 fasteners represent the most critically loaded elements of the assembly, with a maximum stress of 146.62 [MPa] and a safety factor of 1.88 obtained on a refined mesh. While this value falls below the 3.0 target applied to the bulkhead plate itself, it remains acceptable given the conservative assumptions used in the simulation. The minimum stress on the plate is 4.67 MPa, confirming that material is concentrated in the loaded regions while the rest of the plate carries negligible load.

SUSTAINER - Avionic bay top bulkhead

A finite element analysis was performed in Siemens NX Simulation with boundary conditions applied to the M5 screw holes and the recovery load applied at the U-bolt hole locations. Two critical regions were identified. The U-bolt holes experience a maximum stress of 46.43 MPa, well above the minimum safety factor of 3.0 targeted for recovery-critical structural components. The axial M5 screw holes represent the most critically loaded region, with a maximum stress of 88.50 MPa, yielding a safety factor of 3.11 against the 6061-T6 aluminum yield strength of 275 [MPa], which remains acceptable for the design.

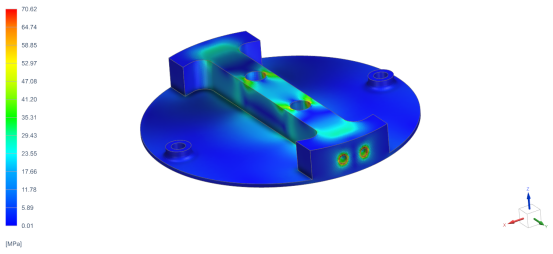


Fig. 74 Isometric view of the sustainer's bottom avionic bay bulkhead finite element analysis

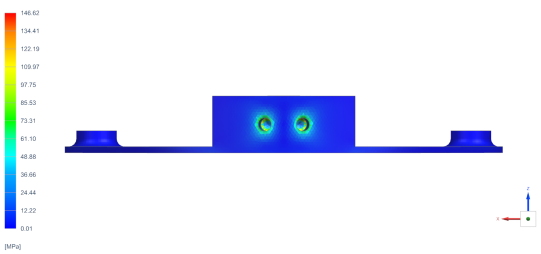


Fig. 75 Front view of the sustainer's bottom avionic bay bulkhead finite element analysis

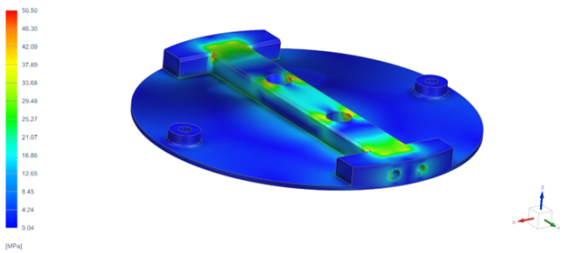


Fig. 76 Isometric view of the sustainer's top avionic bay bulkhead finite element analysis

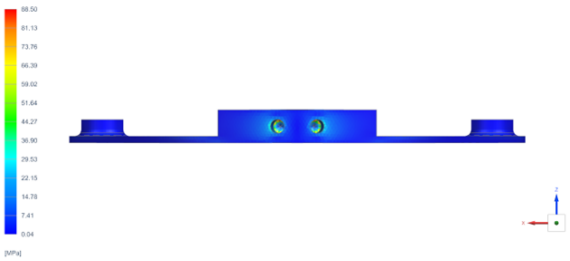


Fig. 77 Front view of the sustainer's top avionic bay bulkhead finite element analysis

Appendix L - Composite Manufacturing

All tubes are student-designed and student-manufactured (SRAD) using a vacuum-assisted resin infusion process on a reusable 142 mm outer-diameter aluminum mandrel. A mylar release sheet is applied to the mandrel prior to layup to ensure clean demoulding without surface damage.

Dry fibre layup: Dry fabric plies are roll-wrapped tightly onto the mandrel using 3M spray adhesive to hold each ply in place. This dry application method ensures a uniform, wrinkle-free layup and eliminates the veining defects commonly introduced when a wet-wound tube is consolidated under vacuum bag pressure.

Resin infusion: Once the desired ply count is reached, a peel ply and infusion mesh are applied over the stack to promote uniform resin flow. Resin feed lines are distributed along the part, with a resin intake puck at one end and a vacuum puck at the other. The assembly is sealed in a vacuum bag with sealant tape, and resin is drawn through the dry preform by vacuum pressure alone. This simultaneously degasses the resin, reducing porosity and improving fibre-to-resin ratio consistency and strength compared to wet layup. Infusion continues for approximately 30 minutes until resin breakthrough is observed at the vacuum puck.

Resin quantity calculation: The resin mass required to achieve the target 60/40 fibre-to-resin ratio is calculated prior to each infusion, ensuring full wetout without excess matrix mass. For a tube of length L and inner diameter d, with n plies of areal weight A_f:

$$m_{\text{resin}} = \left(\frac{n \cdot A_f}{1.5} + m_{\text{losses}} \right) \cdot L \cdot \pi d \tag{35}$$

where the factor of 1.5 follows from the 60/40 ratio ($m_{\text{resin}} = 0.4/0.6 \cdot m_{\text{fibre}}$), and m_{losses} accounts for resin absorbed by the peel ply and feed lines (700 g/m²). For example, for the 142 mm carbon fibre sustainer motor tube ($n = 5$, $A_f = 205$ g/m², $L = 1$ m):

$$\begin{aligned} m_{\text{resin}} &= \left(\frac{5 \times 205}{1.5} + 700 \right) \times 1 \times \pi \times 0.142 \\ &= 1383\,\text{g/m}^2 \times 0.446\,\text{m}^2 = 617\,\text{g} \end{aligned} \tag{36}$$

The West System epoxy used requires a 5:1 epoxy-to-hardener mix ratio by mass, applied to the total resin mass above.

Cure and finishing: The bagged assembly is oven-cured at 75 °C for 90 minutes. After demoulding, the peel ply is removed, the tube is cut to the required length, and the outer surface is sanded to the specified finish.

Appendix M - Recovery Schematics

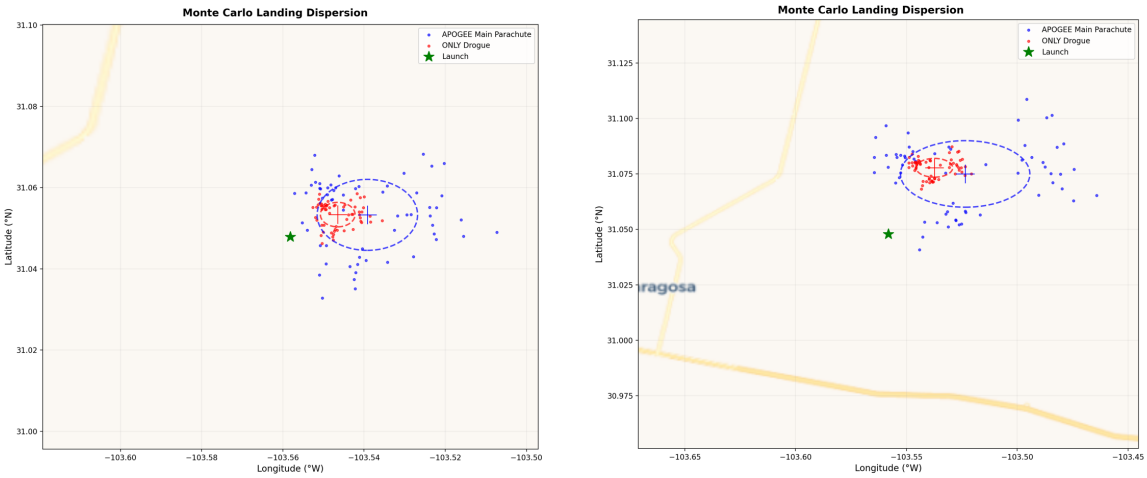


Fig. 78 Monte-Carlo drift simulations for the sustainer (to the right) and booster (to the left) to statistically predict the landing zone of the 2 stages of the rocket using historical multi level wind profile of the IREC launch site. Those simulations show the average landing site of the 2 stages for the different recovery scenario possible, including ballistic, nominal and main at apogee. Those simulations are done on our SRAD simulator called SimulOutil using the parachutes chosen in the design section, the rocket geometry, the launch site conditions (altitude, temperature, location, rail length, rail angle, rail direction) and most importantly historical data/model of June wind profiles.

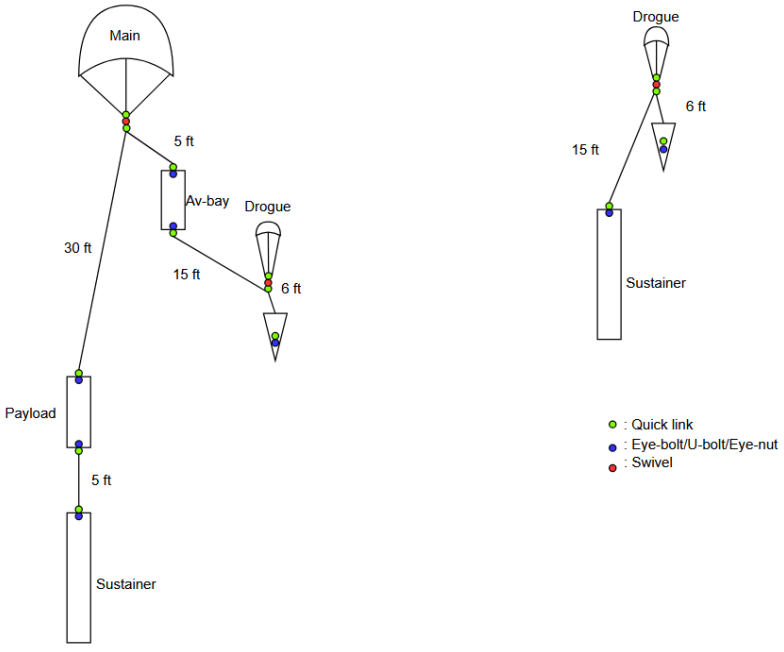


Fig. 79 Recovery line diagram for the sustainer. Drogue parachute deployment state on the right and main parachute deployment state on the left.

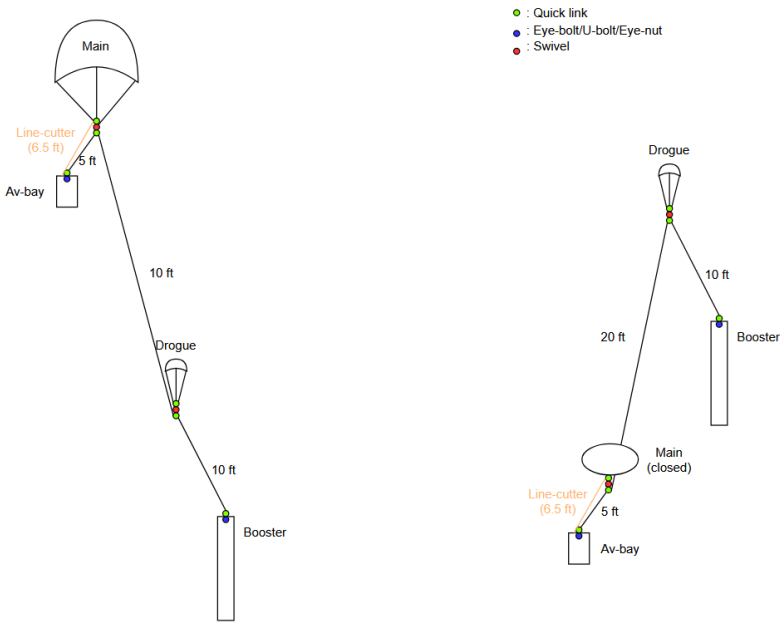


Fig. 80 Recovery line diagram for the booster. Drogue parachute deployment state on the right and main parachute deployment state on the left.

Appendix N - Load Distribution in a Two-Anchor Rope System Under Parachute Drag Force

Load Distribution in a Two-Anchor Rope System Under Parachute Drag Force

General Derivation for Elastic and Inertial Contributions

System Description

Consider a continuous rope of cross-sectional area A and elastic modulus E, attached at its two ends to structural anchors (bulkheads). A parachute is attached at an intermediate point along the rope, dividing it into two segments:

- **Segment A**: length L_A, connecting the attachment node to Bulkhead A, which carries a concentrated mass m_A.
- **Segment B**: length L_B, connecting the attachment node to Bulkhead B, which carries a concentrated mass m_B.

The parachute exerts a total drag force F at the attachment node, directed along the rope axis (both segments align with the force direction, as a flexible rope cannot sustain transverse loads in equilibrium).

Mechanical Model

Each rope segment is modelled as a linear axial spring with stiffness:

$$k_i = \frac{EA}{L_i}, \qquad i \in \{A,\, B\} \tag{1}$$

The two masses m_A and m_B are rigidly attached at the far ends. Letting u_A, u_B, and u_n denote the axial displacements of mass A, mass B, and the node, respectively, the elongations of the two segments are:

$$\delta_A = u_A - u_n, \qquad \delta_B = u_B - u_n \tag{2}$$

Equations of Motion

Applying Newton's second law to each mass and enforcing equilibrium at the massless node:

$$m_A\,\ddot{u}_A = -k_A\,(u_A - u_n) \tag{3}$$

$$m_B\,\ddot{u}_B = -k_B\,(u_B - u_n) \tag{4}$$

$$k_A\,(u_A - u_n) + k_B\,(u_B - u_n) = F(t) \tag{5}$$

Elastic (Quasi-Static) Contribution

When the load is applied slowly relative to the system's natural period, inertial effects are negligible. The node displacement adjusts such that both segments share the load in proportion to their stiffness. From equation (5) with $\ddot{u}_A = \ddot{u}_B = 0$:

$$T_A^{\text{elastic}} + T_B^{\text{elastic}} = F \tag{6}$$

The compatibility condition (both segments undergo the same node displacement) gives:

$$\frac{T_A^{\text{elastic}}}{k_A} = \frac{T_B^{\text{elastic}}}{k_B} \quad \Longrightarrow \quad \frac{T_A^{\text{elastic}}}{T_B^{\text{elastic}}} = \frac{k_A}{k_B} = \frac{L_B}{L_A} \tag{7}$$

Solving simultaneously:

$$\boxed{T_A^{\text{elastic}} = F\,\frac{k_A}{k_A + k_B} = F\,\frac{L_B}{L_A + L_B}} \tag{8}$$

$$T_B^{\text{elastic}} = F\,\frac{L_A}{L_A + L_B} \tag{9}$$

Note that the *shorter* segment carries the *larger* share of the load, as it is stiffer.

Inertial Contribution

When the load is applied as a brief impulse (duration t_{shock} much shorter than the wave transit time L/c, where $c = \sqrt{E/\rho}$ is the axial wave speed), the rope has no time to deform differentially. The system behaves as a rigid body: both masses experience the same instantaneous acceleration a.

From Newton's second law applied to the total system:

$$F = (m_A + m_B)\,a \quad \Longrightarrow \quad a = \frac{F}{m_A + m_B} \tag{10}$$

The tension in each segment equals the force required to accelerate the mass at its far end:

$$\boxed{T_A^{\text{inertial}} = m_A\,a = F\,\frac{m_A}{m_A + m_B}} \tag{11}$$

$$T_B^{\text{inertial}} = F\,\frac{m_B}{m_A + m_B} \tag{12}$$

General Combined Formula

The physical regime is characterised by the dimensionless coupling parameter α, defined as the ratio of the shock duration to the axial wave transit time:

$$\alpha = \frac{t_{\text{shock}}}{L/c} \in [0,\, 1] \tag{13}$$

The tension in each segment is obtained by linearly interpolating between the two limiting cases:

$$\boxed{T_A = F\left[\alpha\,\frac{m_A}{m_A + m_B} + (1-\alpha)\,\frac{L_B}{L_A + L_B}\right]} \tag{14}$$

$$T_B = F\left[\alpha\,\frac{m_B}{m_A + m_B} + (1-\alpha)\,\frac{L_A}{L_A + L_B}\right] \tag{15}$$

Verification: $T_A + T_B = F\,[\alpha + (1-\alpha)] = F$ ✓

α	Physical regime	Governing mechanism
$\alpha \to 0$	Quasi-static loading	Rope stiffness (segment lengths)
$\alpha \to 1$	Impulsive shock	Mass distribution
$0 < \alpha < 1$	Transient dynamic	Both contributions

Conservative Design Case

For parachute deployment, the shock duration is on the order of milliseconds, while the axial wave transit time in a short nylon or Kevlar rope is of the same order of magnitude. In the absence of measured shock duration data, the conservative design assumption is $\alpha = 1$, which eliminates the geometric contribution entirely:

$$\boxed{T_A^{\text{design}} = F\,\frac{m_A}{m_A + m_B}, \qquad T_B^{\text{design}} = F\,\frac{m_B}{m_A + m_B}} \tag{16}$$

Each bulkhead and its associated U-bolt should be sized to withstand the tension given by equation (16), with appropriate safety factors applied per the applicable structural standard.

Justification of $\alpha = 1$ for Kevlar shock cords

The validity of the $\alpha = 1$ assumption can be verified explicitly by comparing the axial stress wave transit time in the rope to the duration of the parachute opening shock.

Axial wave speed in Kevlar

The speed at which a stress wave propagates along a rope is:

$$c = \sqrt{\frac{E}{\rho}} \tag{17}$$

For Kevlar, the elastic modulus is $E \approx 70$–125 GPa and the density is $\rho \approx 1440$ kg/m^3, giving:

$$c = \sqrt{\frac{70 \times 10^9}{1440}} \approx 7000 \text{ m/s} \tag{18}$$

Wave transit time

The time required for the stress wave to travel the full rope length $L = L_A + L_B$ is:

$$t_{\text{propagation}} = \frac{L}{c} \tag{19}$$

For a representative rope length of $L = 2$ m:

$$t_{\text{propagation}} = \frac{2}{7000} \approx 0.29 \text{ ms} \tag{20}$$

Opening shock duration

The duration of a parachute opening shock depends on deployment speed and system compliance. For small parachutes, values reported in the literature range from $t_{\text{shock}} \approx 5$ to 50 ms.

Resulting value of α

$$\alpha = \frac{t_{\text{shock}}}{t_{\text{propagation}}} = \frac{5 \text{ ms}}{0.29 \text{ ms}} \approx 17 \gg 1 \tag{21}$$

Since α is defined on $[0, 1]$ and the computed ratio far exceeds unity, it is clamped to $\alpha = 1$. This result holds across the full range of expected shock durations: even at $t_{\text{shock}} = 5$ ms (the shortest plausible value), the stress wave completes approximately 17 round trips along the rope before the load peaks. The rope therefore reaches mechanical equilibrium nearly instantaneously relative to the shock timescale.

For a Kevlar shock cord of any length up to several metres, the condition $t_{\text{shock}} \gg t_{\text{propagation}}$ is satisfied for all realistic parachute opening scenarios. The elastic (geometric) contribution in equation (14) is therefore negligible, and the conservative design equations (16) are not merely a safe upper bound, they are physically accurate.

Appendix O - Altimeter configurations

Featherweight Blue Raven 1 – Sustainer

Event Name	Trigger Type	Trigger Setting	Output Channel
Sustainer Motor Ignition	Timer	8 sec after liftoff	3rd
Apogee Drogue Deployment	Barometric Apogee Detection	–	Apo
Main Parachute Deployment	Barometric Altitude Detection	1500 ft	Main

Table 31: Flight Event Configuration - Blue Raven Sustainer

Parameter	Value	Units and Typical Use Case
AGL1 (Main chute deployment altitude)	< 1500 ft	Barometric altitude for main parachute deployment above pad
AGL2 (Staging ignition altitude)	> 4500 ft	Barometric altitude for altitude-based staging ignition
FANG1 (Current tilt angle limit)	$< 20.0°$	Max allowable current tilt angle during nominal ascent
FANG2 (End of ascent detection)	$> 20.0°$	Tilt angle threshold to detect end of ascent
FUTANG (Future Tilt Angle)	$> 15.0°$	Expected tilt trigger after 3 seconds
NOMANG (Max Nominal Ascent Tilt)	$< 60.0°$	Max tilt angle during ascent; false after violation
TVAL1 (Inhibit output after)	< 8.2 sec	Prevent outputs too long after liftoff
TVAL2 (Enable output after)	$> 7.8sec$	Allow outputs after this time delay from liftoff
VEL1 (Transonic inhibit velocity)	< 400 ft/sec	Prevent deployment during transonic glitches
VEL2 (Minimum stage ignition velocity)	> 300 ft/sec	Inhibit staging if rocket is going too slowly
BVEL (Barometric downward velocity)	< -200 ft/sec	Early deployment detection if falling too fast
BURNTCOUNT (Motor Burnout Counter)	> 1	Burnout counter increments after deceleration; used for staging logic

Table 32: Flight Parameters and Thresholds - Blue Raven Sustainer

Stratologger CF - Sustainer
The StratoLogger CF provides backup deployment control for the sustainer stage, ensuring recovery redundancy.
Programming Configuration:

- Apogee event: deploy backup drogue at apogee with 1 second delay.
- Main event: deploy backup Main at 1500 ft with 1 second delay.

Featherweight Blue Jay Plus – Booster

Event Name	Trigger Type	Trigger setting	Output Channel
Booster Redundant Apogee Deployment	Barometric Apogee Detection	–	Apo
Booster Redundant Main Chute Deployment	Barometric Altitude Detection	1500 ft	Main

Table 33: Blue Jay Plus Programming Settings for Booster

Description	Value	Units and Use Case
Barometric altitude above the pad (main chute deployment)	$<$ 1500 ft	Barometric altitude for main chute deployment
Barometric altitude above the pad (staging ignition)	$>$ 100 ft	Barometric altitude for altitude-based staging ignition
Tilt angle (detected end of ascent)	$>$ 90°	Detect end of ascent
Time since liftoff (inhibit output)	$<$ 1 s	Prevent outputs too soon after liftoff
Time since liftoff (enable outputs)	$>$ 5.0 s	Allow outputs after this time delay
Accel-based total velocity (transonic inhibit)	$<$ 2000 ft/s	Prevent deployment during transonic glitches
Barometric downward velocity (early deployment detection)	$<$ -200 ft/s	Early main deployment if falling too fast
Motor Burnout Counter	$>$ 1	Detect motor burnout by changes in velocity

Table 34: Blue Jay Plus Flight Settings Parameters

Featherweight Blue Raven 2 – Booster

Event Name	Trigger Type	Trigger Setting	Output Channel
Stage separation	Timer	5.1 sec after liftoff	3rd
Apogee Drogue Deployment	Barometric Apogee Detection	–	Apo
Main Parachute Deployment	Barometric Altitude Detection	1500 ft	Main

Table 35: Flight Event Configuration - Blue Raven Booster

Parameter	Value	Units and Typical Use Case
AGL1 (Main chute deployment altitude)	< 1500 ft	Barometric altitude for main parachute deployment above pad
AGL2 (Stage separation altitude)	> 2500 ft	Barometric altitude for altitude-based staging ignition
FANG1 (Current tilt angle limit)	$< 20.0^\circ$	Max allowable current tilt angle during nominal ascent
FANG2 (End of ascent detection)	$> 20.0^\circ$	Tilt angle threshold to detect end of ascent
FUTANG (Future Tilt Angle)	$> 15.0^\circ$	Expected tilt trigger after 3 seconds
NOMANG (Max Nominal Ascent Tilt)	$< 60.0^\circ$	Max tilt angle during ascent; false after violation
TVAL1 (Inhibit output before)	< 5.0 sec	Prevent outputs too soon after liftoff
TVAL2 (Enable output after)	> 5.0 sec	Allow outputs after this time delay from liftoff
VEL1 (Transonic inhibit velocity)	< 400 ft/sec	Prevent deployment during transonic glitches
VEL2 (Minimum stage ignition velocity)	> 300 ft/sec	Inhibit staging if rocket is going too slowly
BVEL (Barometric downward velocity)	< -200 ft/sec	Early deployment detection if falling too fast
BURNTCOUNT (Motor Burnout Counter)	> 1	Burnout counter increments after deceleration; used for staging logic

Table 36: Flight Parameters and Thresholds - Blue Raven Booster

Appendix P - Avionics Wiring Diagram

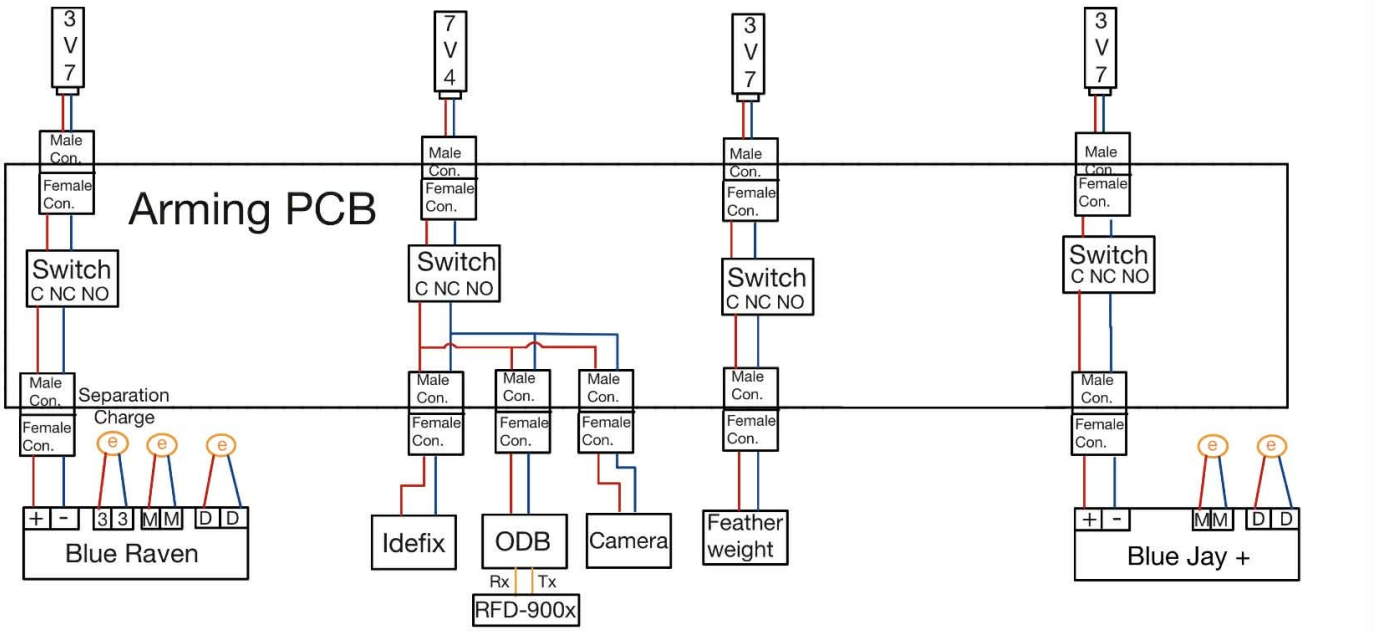


Fig. 81 Wiring diagram for the avionics bay of the booster.

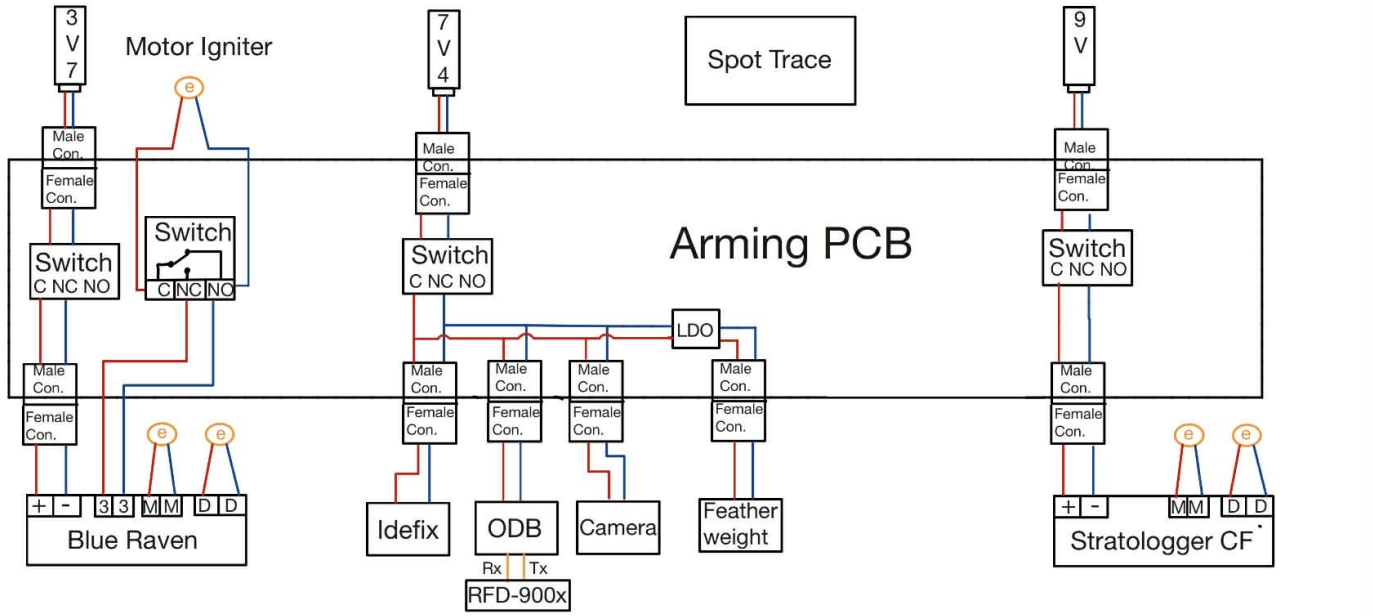


Fig. 82 Wiring diagram for the avionics bay of the sustainer.